Tail Certificates and Achievement-Set Geometry

for Erdős Problems 249 and 257

Will Cook July 2026

Abstract

For the totient constant $S = \sum_{n\geq 1} \varphi(n)/2^n$, one certified Farey window excludes every rational representation $S = p/q$ with $q \leq Q_0 \approx 7.96 \times 10^{34}$. We also prove that irrationality of S is equivalent to non-integrality of every positive binary tail difference, with a complete finite certificate at each fixed pair of parameters; one integral difference would instead force rationality. The missing input is an unbounded supply of these certificates, and the checks through $t = 64$ are finite evidence only. For Erdős #257, we formalise Erdős's classical full-support theorem in every integer base and several structured-support cases. At base 2, the Mersenne achievement set is compact, perfect, nowhere dense, and of measure one. Any infinite rational-valued support must have unbounded positive carry states and sublogarithmic divisor-coverage gaps. Membership of 1/2, which would refute the universal statement, is equivalent to infinitely many greedy skips. The upper transition and the middle coordinate -3 cannot be final. To exclude a final skip altogether, it remains to rule out the exceptional coordinates $-2, -1$ and prove the exact future-tail inequality for every other non--3 middle transition. Neither problem is settled.

1 Introduction

The two questions are whether $S = \sum_{n\geq 1} \varphi(n)/2^n$ is irrational (#249), and whether $\sum_{n\in A}(2^n - 1)^{-1}$ is irrational for every infinite $A \subseteq \mathbb{N}$ (#257). The formulations occur in the Erdős–Graham collection and Erdős's later survey [3, 4]; the numbering follows the maintained catalogue [21].

For #257, proving irrationality for every infinite support is the universal direction; constructing an infinite support of sum 1/2 would refute it. The results below contain unconditional positive cases and an exact conditional counterexample route, but they reach neither endpoint. For #249, and for the half-value counterexample branch of #257, tail and carry identities separate exact finite arithmetic from one analytic or combinatorial cofinality statement. The universal #257 direction remains open for arbitrary infinite supports. The status classes are:

status	Erdős #249	Erdős #257
Classical	Lambert-series and convolution identities used to represent S.	Full support in every base $b \geq 2$; pairwise-coprime and periodic support families.
Unconditional/exact	One window gives $q > Q_0$ for every rational $S = p/q$; irrationality is equivalent to non-integrality of every positive tail difference and to pointwise finite certificates.	Measure-one nowhere-dense achievement geometry; exact rational carry certificates; unbounded states and sublogarithmic zero gaps over infinite rational supports; exact classification of 1/2.
Verified finite	28 diagonal certificates through $t = 64$ and further bounded exact verifications.	Local band and window identities, bounded reset rows, and explicit finite counterexamples.
Conditional	A cofinal certificate or scalar-slack supply implies irrationality.	Cofinal seam, window, strip, or terminal hypotheses imply an infinite support of sum 1/2.
Open	An unbounded certificate supply.	Universal direction: all infinite supports. Counterexample direction: cofinal false terminals; concretely, the two final-skip obligations in Section 4.

Reading map. Section 2 states the two exact proof spines and their unresolved hypotheses. Section 3 gives the common Mersenne–Lambert transform; Sections 4 and 5 give the theorem-level detail. Section 8 states the remaining questions. Appendices A and B contain the local half-value and rational-carry machinery; the later appendices contain the totient-certificate reductions and the independent Lambert, probability, and finite-algebraic structure. Linked mathematical phrases open exact Lean proofs. The companion develops the affine and first-harmonic #249 theory without changing its open status.

Keywords. irrationality; Erdős–Borwein constant; Euler totient; Lambert series; achievement sets; binary carry systems; Lean 4. **MSC 2020.** 11J72 (primary); 11A25, 68V20 (secondary).

2 The two exact spines

2.1 Erdős #249: an exact certificate normal form

Write

$$R_N = \sum_{m \geq 1} \frac{\varphi(N+m)}{2^m}.$$

The formal source proves the exact chain

$$\begin{aligned} S \notin \mathbb{Q} \quad &\iff \quad R_{N+h} - R_N \notin \mathbb{Z} \quad \text{for every } h > 0 \text{ and every } N \\ &\iff \quad \text{for every } h > 0, N \text{ there is an } L \text{ with } \mathsf{certifiedKill}(h, N, L). \end{aligned}$$

Here $\mathsf{certifiedKill}(h, N, L)$ is a finite residue computation with a rigorous tail allowance. The second equivalence is completeness of the certificate: at fixed h, N, some finite depth succeeds exactly when the tail difference is non-integral. Conversely, integrality of one positive-shift difference forces S to be rational.

This normal form is accompanied by an unconditional theorem: if $S = p/q$, then

$$q > Q_0 = 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053.$$

The exact open statement is an unbounded supply of non-integral tail differences, equivalently of finite certificates. Corollary 5.5 gives a sufficient lcm-diagonal form, and finite diagonal instances are verified through $t = 64$, but no unbounded supply is proved. The displayed chain is formalised by the tail-difference equivalence, the pointwise certificate equivalence, and the fixed-parameter certificate-completeness theorem. The unconditional finite bound is the denominator exclusion.

2.2 Erdős #257: the exact half-value counterexample spine

Let

$$\mathcal{A} = \left\{ \sum_{n \in A} \frac{1}{2^n - 1} : A \subseteq \mathbb{N},\ 0 \notin A \right\}.$$

The weights are strictly superincreasing, so the normalized support of a value is unique. For $1/2$, the following are equivalent:

$$\begin{aligned} \frac{1}{2} \in \mathcal{A} \quad &\iff \quad \text{the greedy skipped support is infinite} \\ &\iff \quad \text{false successor terminals occur beyond every bound.} \end{aligned}$$

A finite normalized support cannot have value $1/2$. Thus membership would produce an infinite support of rational sum and refute the universal statement of Erdős #257. The required counterexample condition is therefore exact: cofinally many false terminals, equivalently no last greedy skip.

More generally, rationality over an infinite support forces an unbounded positive integer carry orbit. Further exact consequences are reciprocal-mass lower bounds and sublogarithmic zero-window bounds for the divisor-count coefficient of any rational-valued support. These are unconditional rigidity theorems, not a proof that rational infinite supports cannot exist.

The strongest current last-skip contradiction is conditional. At the seam following a putative last skip, the upper transition is impossible, and among the middle coordinates $-3, -2, -1$, the coordinate -3 is impossible. The remaining proof must exclude the exceptional coordinates $-2, -1$ and establish the exact future-tail inequality at every other non--3 middle transition. Neither assertion is proved for the actual greedy orbit. The membership chain is formalised by the infinite-skip equivalence, unbounded-terminal equivalence, no-last-skip equivalence, and the finite-support exclusion. The conditional residual is the remaining-tail implication. Its local transition exclusions are stated and linked where they are used in Theorem 4.7; Appendix A contains the supporting coordinate analysis.

Prior-art boundary. The full-support theorem is Erdős's, and Borwein proved a broader irrationality theorem containing the same Mersenne–Lambert specialisation [1, 5]. Pairwise-coprime supports are classical [2], and Luca–Tachiya treat periodic rational coefficients by another method [13]. Kovač–Tao supply the strict-tail achievement-set context [14]; Wang–Grau Ribas supply the closest recorded integral carry recurrence [15]. Crandall and Campbell concern adjacent binary-expansion questions [16, 17]. The explicit denominator bound, certificate equivalence, and last-skip reduction have no exact antecedent in the maintained comparison record; that absence is not itself a novelty claim. The theorem-specific comparison record is maintained with the release.

3 The Mersenne–Lambert ladder

Define the Lambert-type transform of an arithmetic function f by

$$L(f) \;=\; \sum_{n\geq 1} \frac{f(n)}{2^n - 1} \;=\; \sum_{m\geq 1} \frac{(f * \mathbf{1})(m)}{2^m}, \qquad (f * \mathbf{1})(m) = \sum_{d\mid m} f(d),$$

the second equality being the standard Lambert rearrangement (one Dirichlet convolution by the constant $\mathbf{1} = \zeta$); Apostol supplies the classical arithmetic-function and Dirichlet-convolution background [6]; Merca introduced a modern Lambert-series factorisation theorem [7], and Merca–Schmidt give a synthesis/unification treatment [8]. Walking the ladder by convolution gives five values with rational, irrational, open, or transcendental status.

input f	value	mathematical status	treatment here
μ	$L(\mu) = 1/2$	rational	formalised
φ	$L(\varphi) = 2$	rational	formalised
$\mathbf{1}$	$L(\mathbf{1}) = E$	irrational	formalised
$A = \varphi * \mu$	$L(A) = S$	open (#249)	identities and bounds formalised
Id	$L(\mathrm{Id}) = \sum \sigma(m)/2^m$	transcendental	cited [11]

The weight $A = \varphi * \mu$ is the primitive-conductor count: $A(p) = p - 2$ on primes, $A \geq 0$, and $A * \zeta = \varphi$. One convolution by ζ separates the open constant S from the rational 2, and one more separates rational from transcendental ($\varphi * \zeta = \mathrm{Id}$, $\mathrm{Id} * \zeta = \sigma$). So S is the rung one convolution from the machine-checked irrational E: the same series $\sum(\cdot)/(2^d - 1)$, with the constant weight 1 replaced by the nonnegative weight A.

Proposition 3.1 (the rational rungs)**.** *$\sum_{d\geq 1} \mu(d)/(2^d - 1) = 1/2$ and $\sum_{d\geq 1} \varphi(d)/(2^d - 1) = 2$, exactly.*

Formalised as the Möbius rung and the totient rung.

Proposition 3.2 (the positive lift, $L(A) = S$). *With the primitive-conductor weight $A = \varphi * \mu$,*

$$\sum_{d \geq 1} \frac{A(d)}{2^d - 1} = \sum_{n \geq 1} \frac{\varphi(n)}{2^n} = S.$$

Formalised as the positive-lift identity. This places S in the same $\sum(\cdot)/(2^d - 1)$ family as E, with a nonnegative weight.

Probabilistic reading. If X, Y are independent fair-coin waiting times, $\Pr(X = n) = 2^{-n}$, then

$$S = \tfrac{1}{2} + \Pr(\gcd(X, Y) = 1).$$

Thus Erdős #249 also asks whether the probability that two independent fair-coin waiting times are coprime is irrational. The exact identity is Proposition 5.2; its Möbius-square form and the periodic and finite-algebraic variants of the ladder are collected in Section D.1.

Section consequence and residual. The exact contribution of this section is the coordinate change $L(A) = S$, together with the two rational rungs above it in the same Lambert family. These identities explain why the Mersenne–Lambert coordinates are natural for both problems, but they do not imply that S is irrational. For #249 the remaining issue is still the unbounded certificate supply isolated in Section 5; for #257 the ladder supplies context rather than either the universal theorem or a rational counterexample.

4 Erdős–Borwein-type irrationality (the #257 direction)

There are two logically distinct directions. The universal direction asks for irrationality for every infinite support; the counterexample direction asks for one infinite support with rational value, and the target 1/2 gives an exact formulation of that problem. This section first states the classical and structured irrationality theorems, then gives the base-2 achievement-set geometry and the constraints forced by a hypothetical rational value. It ends with the exact classification of the target 1/2 and the exact residual condition left by the current final-skip argument. Local rewind, window, cylinder, and finite-stage results are collected in Appendix A.

4.1 Full support: the Erdős–Borwein constant, every base

Theorem 4.1 (full-support irrationality). *For every integer $b \geq 2$, the series $\sum_{n \geq 1} \frac{1}{b^n - 1}$ is irrational. In particular the Erdős–Borwein constant $E = \sum_{n \geq 1} 1/(2^n - 1)$ is irrational.*

The all-base theorem and its base-2 corollary are formalised. This is Erdős's 1948 theorem [1]. In the ladder of Section 3 it is the rung $L(\mathbf{1})$, and it is the fixed machine-checked point one convolution away from the open S.

The proof begins with the Lambert identity

$$\sum_{n \geq 1} \frac{1}{b^n - 1} = \sum_{m \geq 1} \frac{\tau(m)}{b^m},$$

where $\tau(m)$ is the number of positive divisors of m; see the Lambert identity. If the right-hand side were rational with reduced denominator q, then a non-integral integer multiple could not lie at distance less than $1/q$ from an integer. It is therefore enough, for every q, to construct N

such that $b^N \sum_m \tau(m)b^{-m}$ is non-integral but lies within $1/q$ of an integer. This is the formal near-integer criterion.

The construction of N is divided into three finite parts. First, a bounded Bertrand argument supplies disjoint prime blocks, and the Chinese remainder theorem chooses an arithmetic progression on which

$$b^r \mid \tau(N+r) \qquad (1 \le r \le K).$$

These divisibilities make the first K terms of the shifted tail integral. The multiplicativity step that converts prescribed prime valuations into the displayed divisibility is the prime-block divisibility lemma. Second, divisor pairing bounds the average of the weighted middle window

$$\sum_{K<r\le L} \tau(N+r)b^{L-r}$$

along that progression. A pigeonhole choice then selects one translate with a small middle contribution; see the weighted-middle selection lemma. Third, the elementary estimate $\tau(n) \le n$ controls the infinite tail after L. The parameters are chosen so that the middle and far-tail bounds together are smaller than $1/q$; the shifted tail is strictly positive because every divisor count is positive, so the resulting dilation is not an integer.

Thus the analytic series is consumed only at the Lambert identity and the geometric tail estimate. The remaining proof is a finite CRT construction, divisor counting, and explicit parameter arithmetic. The final certificate existence theorem holds for every $b \ge 2$ and every precision q; see the uniform certificate theorem. Its composition with the near-integer criterion is the theorem cited above. This is a formalisation of the classical full-support theorem, not a new irrationality result.

4.2 Named infinite-support cases

Beyond full support, the development proves irrationality for a family of infinite supports A, at every base $b \ge 2$. Each is a formalised case of the #257 statement family. None is the universal statement, and none is claimed as new mathematics.

Theorem 4.2 (support-class family). *For every integer $b \ge 2$, the series $\sum_{n\in A} 1/(b^n-1)$ is irrational for each of the following supports A:*

(a) *factorial support $A = \{n! : n \ge 1\}$, giving $\sum 1/(b^{n!}-1)$;*

(b) *power-of-two support $A = \{2^n : n \ge 0\}$, giving $\sum 1/(b^{2^n}-1)$;*

(c) *multiples $A = d\mathbb{N}$;*

(d) *pairwise-coprime supports with summable reciprocals (Erdős's condition [2]);*

(e) *eventually-periodic supports, residue classes, and the odd numbers.*

These sort into three groups. The classical case is (d): infinite pairwise-coprime A with $\sum_{a\in A} 1/a < \infty$, formalised from Erdős's condition. The lcm-gap named cases are (a) and (b), whose support gaps outgrow the running lcm; the structured families are (c) and (e).[1] Formalised case by case by the factorial-support theorem, power-of-two-support theorem, multiple-support theorem, pairwise-coprime-support theorem, and eventually-periodic-support theorem. The residue-class and odd supports in part (e) are direct specialisations of the last theorem; the first two theorems already hold for every base $b \ge 2$.

[1] Luca–Tachiya prove the broader eventual-periodic rational-coefficient theorem; its indicator specialisation directly contains case (e) [13]. Their Chowla–Erdős-style large-modulus proof is distinct from the periodic-divisor certificate route formalised here, and the full mixed-sign coefficient theorem is not claimed.

The prime support is now known unconditionally: Tao and Teräväinen proved that

$$\sum_{p} \frac{1}{2^p - 1} = \sum_{n \geq 1} \frac{\omega(n)}{2^n}$$

is irrational [18]. Their proof uses quantitative two-point correlations of multiplicative functions and is not formalised here. This settles an important special case, while the universal support statement remains open.

The factorial and power-of-two cases are worth a word because they are not reachable by a Liouville shortcut: the power-of-two gap $2^k - \mathrm{lcm}$ stays within a bounded ratio of the lcm, so the terms do not decay fast enough for a transcendence-measure argument, and a uniform criterion is needed. The mechanism is *period noncollapse*: a prime-valuation-deficit witness forces the multiplicative order of the base modulo the relevant modulus not to collapse, which rules out the long repeated digit blocks a rational value would require.

4.3 The base-2 achievement set

For $n \geq 1$, put

$$w_n = \frac{1}{2^n - 1}, \qquad T_n = \sum_{m > n} w_m, \qquad \mathcal{A} = \left\{ \sum_{n \in A} w_n : A \subseteq \mathbb{N},\ 0 \notin A \right\}.$$

For $x \geq 0$, define the greedy residuals by $r_0(x) = x$ and

$$\varepsilon_n(x) = \begin{cases} 1, & w_n \leq r_{n-1}(x), \\ 0, & w_n > r_{n-1}(x), \end{cases} \qquad r_n(x) = r_{n-1}(x) - \varepsilon_n(x) w_n.$$

Theorem 4.3 (achievement-set geometry and greedy coding)**.** *The following statements hold.*

(i) *$T_n < w_n$ for every $n \geq 1$. Hence every $x \in \mathcal{A}$ has a unique normalized support.*

(ii) *A real number x lies in $\mathcal{A}$ if and only if $x \geq 0$ and $r_n(x) \leq T_n$ for every $n \geq 0$. In that case $x = \sum_{n \geq 1} \varepsilon_n(x) w_n$.*

(iii) *The set $\mathcal{A}$ is compact, perfect, totally disconnected, and nowhere dense, and its Lebesgue measure is 1.*

(iv) *Exact rational evaluation of the greedy recurrence is sound. In particular, if an exact residual exceeds an exact upper bound for the remaining tail at some finite level, then $x \notin \mathcal{A}$.*

The strict-tail inequality and greedy criterion are the strict-tail theorem and greedy membership theorem. Uniqueness is the support-coding theorem; the four topological and measure properties in part (iii) are proved and linked together in Proposition B.13. Kovač–Tao provide the strict-tail Cantor-set context [14]; no novelty claim is made for that geometry. Exact rational death certificates are given in Appendix B.

Consequence for the half-value branch. Parts (i), (ii), and (iv) make the greedy code a canonical decision procedure, while compactness in part (iii) allows cofinal finite approximations to pass to a point of $\mathcal{A}$. The measure-one statement is global: it does not decide whether the particular point $1/2$ belongs to $\mathcal{A}$. That pointwise question is the separate reduction below.

4.4 Rigidity of a hypothetical rational support

For $A \subseteq \mathbb{N} \setminus \{0\}$, define

$$f_A(m) = \#\{a \in A : a \mid m\}, \qquad X_A = \sum_{a \in A} \frac{1}{2^a - 1} = \sum_{m \geq 1} \frac{f_A(m)}{2^m}, \qquad T_f(N) = \sum_{j \geq 1} \frac{f(N+j)}{2^j}.$$

Theorem 4.4 (rational-support rigidity)**.** *Suppose that* $A \neq \varnothing$ *and*

$$X_A = \frac{p}{2^c v}, \qquad p \in \mathbb{Z}, \quad c \in \mathbb{N}, \quad v \geq 1 \text{ odd.}$$

Then $u_N = vT_{f_A}(c+N)$ *is a positive integer for every* N*, and*

$$u_{N+1} + vf_A(c+N+1) = 2u_N. \tag{4.1}$$

Moreover:

(i) *if* A *is infinite, then* (u_N) *is unbounded;*

(ii) *for every* $\varepsilon > 0$ *there is* $B = B(\varepsilon, c, v)$ *such that every interval* $\{c+N+1, \ldots, c+N+h\}$ *on which* f_A *vanishes satisfies*

$$h \leq \varepsilon \log_2(N+1) + B;$$

(iii) *if* $\rho(A) = \sum_{a \in A} a^{-1} < \infty$*,* $v > 1$*, and* $h = \operatorname{ord}_v(2)$*, then*

$$\rho(A) = \frac{w}{h} + \lim_{M \to \infty} \frac{1}{M} \sum_{N < M} e_N \geq \frac{w}{h},$$

where w *is the number of wraps in one doubling-residue cycle and* $u_N = ve_N + (p2^N \bmod v)$*. If* $(p, v) = 1$*, then* $w \geq 1$*. For* $v = 1$ *and* A *infinite, either* $\rho(A)$ *diverges or* $\rho(A) > 1$*.*

Equation (4.1) is the exact carry recurrence. Part (i) rules out every bounded-state model for an infinite rational support; part (ii) prevents logarithmically long gaps in divisor coverage; part (iii) relates the odd denominator to the reciprocal density of the support. These are necessary consequences of rationality, not a contradiction. The exact tail-orbit statement is Theorem B.1. Parts (i) and (ii) are proved in Theorems B.11 and B.7; the odd-denominator portion of part (iii) is Proposition B.9, and its dyadic endpoint is Corollary B.10. The supporting Boolean-certificate coordinates are developed in Appendix B.

4.5 The target $1/2$

Let

$$\mathcal{S}(x) = \{n \geq 1 : \varepsilon_n(x) = 0\}$$

be the set of exponents omitted by the greedy expansion.

Definition 4.5 (finite greedy seam and terminal bit)**.** For an integer $s \geq 6$, set

$$q_{s,d} = \left\lfloor \frac{4^s}{2^d - 1} \right\rfloor \quad (2 \leq d < s), \qquad K_s = 2^{2s-1} - 2^s.$$

Starting from $R_{s,2} = K_s$, define successively, for $2 \leq d < s$,

$$\beta_{s,d} = \begin{cases} 1, & q_{s,d} \leq R_{s,d}, \\ 0, & q_{s,d} > R_{s,d}, \end{cases} \qquad R_{s,d+1} = R_{s,d} - \beta_{s,d} q_{s,d}.$$

The Boolean word $(\beta_{s,2}, \ldots, \beta_{s,s-1})$ is the *finite greedy seam at row* s, and $\tau_s = \beta_{s,s-1}$ is its terminal bit.

This is an exact integer coordinate model of the real greedy process; Appendix A develops its transition formulas and the comparison with the real residuals.

Theorem 4.6 (exact half-membership classification)**.** *The following are equivalent:*

(i) $\frac{1}{2} \in \mathcal{A}$;

(ii) $\mathcal{S}(1/2)$ *is infinite*;

(iii) $\mathcal{S}(1/2)$ *has no largest element*;

(iv) *for every* N *there is* $s \geq \max\{N, 6\}$ *with* $\tau_s = 0$.

No finite normalized support has value $1/2$. *Consequently, any one of* (i)–(iv) *produces an infinite support* A *with* $X_A = 1/2$*, and therefore refutes the universal form of Erdős #257.*

The greedy-support equivalence is the infinite-skip theorem; the terminal forms are the unbounded-terminal theorem and no-final-skip theorem. Finite supports are excluded by the finite-support theorem. A tempting signed-to-Boolean shortcut also fails: selecting precisely the negative Möbius indices overshoots $1/2$ by at least $1/63$. Proposition B.3 gives the exact sign-separation identity.

Section consequence. Theorem 4.6 replaces a search over arbitrary supports by one canonical quantifier question: does the greedy orbit of $1/2$ skip cofinally? It does not answer that question. The next subsection assumes a possible final skip and identifies exactly what would be required to exclude it.

4.6 The final-skip reduction

For each $s \geq 6$, let
$$G_s = \{d : 2 \leq d < s,\ \beta_{s,d} = 1\}.$$
Let H_s be the unique subset of $\{2, \ldots, s-1\}$ for which $\sum_{d \in H_s} q_{s,d}$ is least among the subset sums strictly exceeding K_s. The transition from row s to row $s+1$ is called *upper*, *middle*, or *right* according as

$$G_{s+1} = H_s, \qquad G_{s+1} = G_s, \qquad G_{s+1} = G_s \cup \{s\},$$

respectively. Thus a final skipped exponent D is an upper or middle transition followed only by right transitions.

At a middle transition $D \geq 13$, put

$$A_D = G_D \cup \{D\}, \qquad P_D = 2f_{G_D}(2D+1) + f_{G_D}(2D+2),$$

and write
$$C_D = 4R_{D,D} - P_D - 4 \in \mathbb{Z}, \qquad \Theta_D = \sum_{j \geq 1} \frac{f_{A_D}(2D+2+j)}{2^j}.$$

Here C_D is the affine carry at the transition and Θ_D is its complete future divisor-incidence tail.

For a finite set $u \subseteq \{1, \ldots, d\}$, write $V(u) = \sum_{n \in u} w_n$. Call (u, d) a *fatal half-gap* if

$$V(u) + T_{d+1} < \frac{1}{2} < V(u) + w_{d+1}.$$

Such a pair certifies that no support agreeing with u through rank d can represent $1/2$.

Theorem 4.7 (final-skip classification and residual inequality)**.** *Nonmembership of* $1/2$ *in* $\mathcal{A}$ *is equivalent to each of the following:*

(i) *the integer seam recurrence is eventually right;*

(ii) *a fatal half-gap exists;*

(iii) *the real greedy expansion of* $1/2$ *has a final skipped exponent.*

At such a final exponent $D \geq 13$*, the upper successor is impossible. On the middle branch the value* $C_D = -3$ *is impossible. The only unresolved exceptional cells for the current two-sided dyadic induction are* $C_D = -2$ *and* $C_D = -1$*.*

If every middle transition at a row $D \geq 13$ *with* $C_D \neq -3$ *satisfies*

$$\Theta_D < C_D, \tag{4.2}$$

then a final skipped exponent cannot exist, and hence $1/2 \in \mathcal{A}$*.*

The three equivalent nonmembership criteria are formalised in fatal-gap classification, nonmembership classification, and final-skip classification. The upper and $C_D = -3$ exclusions are the upper-branch exclusion and middle--3 exclusion. The three exceptional dyadic cells are identified by the three-cell classification. The implication from (4.2) is the tail-dominance theorem. Since $\Theta_D \geq 0$, condition (4.2) would in particular exclude the remaining cells -2 and -1. No proof of (4.2) for the actual greedy orbit is known here.

4.7 What remains open

The two directions remain separate.

(a) The universal problem is to prove $X_A \notin \mathbb{Q}$ for every infinite $A \subseteq \mathbb{N} \setminus \{0\}$. Theorem 4.2 proves several families, and Theorem 4.4 constrains a hypothetical rational support, but neither result supplies the universal quantifier.

(b) The counterexample problem is to prove $1/2 \in \mathcal{A}$. Theorem 4.6 identifies this with infinitely many greedy skips, while Theorem 4.7 reduces the current final-skip argument to two exact obligations: exclude the remaining exceptional cells $C_D = -2, -1$, and prove the tail inequality (4.2) at every other non--3 middle transition. Neither obligation is discharged.

Appendix A records the exact local reductions, finite certificates, counterexamples to stronger local assertions, and closed strategy branches supporting this conclusion. They refine the obstruction; they do not constitute a second theorem spine.

5 The totient constant S (Erdős #249)

Whether

$$S = \sum_{n \geq 1} \frac{\varphi(n)}{2^n}$$

is irrational remains open. The argument has four logically distinct layers. First, a finite Farey computation excludes every rational denominator up to an explicit bound. Second, irrationality is characterised exactly by the existence of finite certificates against every possible eventual binary period. Third, many such certificates are verified at bounded parameters. Fourth, and still missing, is a theorem supplying them at unbounded parameters. Only the fourth layer would settle the problem.

The main section states this theorem spine. Appendix C contains the cone, projection, denominator, parity, and closed-strategy reductions that refine the open fourth layer.

5.1 Unconditional: no small denominator

Theorem 5.1 (denominator exclusion). *Set*

$$Q_0 := 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053 \approx 7.96 \times 10^{34}.$$

For every integer p and every q with $1 \le q \le Q_0$,

$$S \neq \frac{p}{q}.$$

Equivalently, if S is rational then its reduced denominator exceeds that bound.

Formalised as the record denominator exclusion. The bound is reached by a ladder of finite, elementary exclusions,

$$4838 \;\to\; 4\,194\,304 = 2^{22} \;\to\; 2.49 \times 10^{17} \text{ (Farey } K{=}120) \;\to\; 7.96 \times 10^{34} \text{ (Farey } K{=}240),$$

each a mediant/Farey argument that traps any candidate m/q inside a forbidden gap. The mediant framework is historically associated with Farey's 1816 note on vulgar fractions [10]; the finite exclusions themselves are Lean-checked arguments in the formal-source checkpoint. See denominator exclusion through 4838 for the first rung and 120-term Farey bound, 240-term Farey bound for the two Farey windows.

This is an explicit lower bound on the denominator of any rational representation, not a proof of irrationality. A rational with a larger denominator is not excluded by it.

Remark (one bound, four representations). Because the ladder identities of Section 3 are equalities, the same finite record transfers verbatim to three other representations of the constant: to the positive lift $\sum_d A(d)/(2^d-1)$ (positive-lift denominator transfer), to the signed Möbius-square constant $T = \sum_d \mu(d)/(2^d-1)^2$ at half the bound (signed-square denominator transfer), and to the coprimality probability of Section 5.2 at half the bound. Half appears because $S = \frac{1}{2} + T$ turns a denominator d for T into $2d$ for S.

5.2 The geometric picture: S as coprime-pair mass

Let X, Y be independent fair-coin waiting times, $\Pr(X = n) = 2^{-n}$ for $n \ge 1$. Then $\Pr(d \mid X) = 1/(2^d - 1)$, so the Lambert transform is a divisor calculus of X: $L(f) = \mathbb{E}[(f * \zeta)(X)]$. The rungs read as $L(\mu) = \Pr(X = 1) = 1/2$, $L(\varphi) = \mathbb{E}[X] = 2$, $L(\mathbf{1}) = \mathbb{E}[\tau(X)] = E$, and $L(A) = \mathbb{E}[\varphi(X)] = S$.

Counting visible lattice points (a, b) with $a + b = n$, $a > 0$, $\gcd(a, b) = 1$ gives exactly $\varphi(n)$, uniformly in n, so S is itself a coprime-pair mass.

Proposition 5.2 (coprimality probability).

$$S \;=\; \tfrac{1}{2} \;+\; \Pr\big(\gcd(X,Y) = 1\big) \;=\; \tfrac{1}{2} \;+\; \sum_{\substack{a,b \ge 1 \\ \gcd(a,b)=1}} 2^{-(a+b)}.$$

Formalised: coprime-pair representation, with the underlying lattice identity at visible-lattice-point identity. Base 2 is the one point where $\Pr(X = a)\Pr(Y = b) = 2^{-(a+b)}$ needs no normalising constant, so the totient sum and the coprimality probability line up with no boundary correction. In this reading, Erdős #249 asks whether two independent fair-coin waiting times are coprime with irrational probability. After reducing (X, Y) by their gcd, the same distribution has an exact Stern–Brocot cylinder decomposition. Appendix D states the resulting probability law and the sharp Fibonacci stability of its alternating runs. These are unconditional structural theorems about the representation of S; they do not supply the denominator survival needed for irrationality.

5.3 Exact certificate characterisation

The reduction uses only the elementary bound $\varphi(n) \le n$, geometric tail estimates, and finite integer arithmetic. For $t \ge 1$, write $M_t = \operatorname{lcm}(1, \dots, t)$.

Definition 5.3 (tail differences and finite certificates)**.** For $N \ge 0$, define the scaled tail

$$R_N \;=\; \sum_{m \ge 1} \frac{\varphi(N+m)}{2^m}.$$

For $h \ge 1$ and $L \ge 1$, define

$$\Delta_{h,N,L} = \sum_{j=0}^{L-1} \bigl(\varphi(N+h+1+j) - \varphi(N+1+j)\bigr)\, 2^{\,L-1-j} \in \mathbb{Z},$$

and say that (h, N, L) is a *finite certificate* when

$$\mathsf{certifiedKill}(h, N, L): \quad N+h+L+2 \;<\; \bigl(\Delta_{h,N,L} \bmod 2^L\bigr) \;<\; 2^L - (N+h+L+2).$$

The integer $\Delta_{h,N,L}$ contains the first L binary places of $R_{N+h} - R_N$; the two margins bound the omitted tail using $\varphi(n) \le n$. Thus the predicate is decidable by finite integer arithmetic, while its conclusion concerns the infinite tail difference.

Theorem 5.4 (exact tail-period characterisation)**.** *S is irrational if and only if, for every period $h \ge 1$ and every threshold N_0, there exist $N \ge N_0$ and L with $\mathsf{certifiedKill}(h, N, L)$.*

Formalised as the certificate-supply equivalence.

Every multiple of an eventual period is again a period. The least common multiple M_t therefore provides a useful sufficient diagonal form of Theorem 5.4.

Corollary 5.5 (diagonal reduction)**.** *Suppose that for every threshold t_0 there are parameters $t \ge t_0$ and L for which $\mathsf{certifiedKill}(M_t, M_t, L)$ holds. Then S is irrational.*

Formalised: diagonal-supply criterion.

Proposition 5.6 (certificate completeness)**.** *For every $h \ge 1$ and $N \ge 0$,*

$$\bigl(\exists L \ge 1,\ \mathsf{certifiedKill}(h, N, L)\bigr) \quad\Longleftrightarrow\quad R_{N+h} - R_N \notin \mathbb{Z}.$$

Moreover, the following are equivalent:

$$\begin{aligned} &S \notin \mathbb{Q}, \\ &R_{N+h} - R_N \notin \mathbb{Z} && \text{for every } h \ge 1 \text{ and } N \ge 0, \\ &\exists L \ge 1,\ \mathsf{certifiedKill}(h, N, L) && \text{for every } h \ge 1 \text{ and } N \ge 0. \end{aligned}$$

Formalised: certificate-completeness equivalence, all-differences criterion, and pointwise certificate criterion. The converse direction uses the fact that integrality of even one positive-shift tail difference forces S to be rational (integral-difference rationality criterion). Thus the finite predicate is complete at each fixed pair (h, N); the open problem is to prove that the required pairs occur without bound.

Example 5.7 (verified finite range)**.** Lean checks:

- $\mathsf{certifiedKill}(h, 12, 16)$ for every $1 \le h \le 8$;
- a fixed-window certificate for every $1 \le h \le 16$;

- diagonal certificates $\mathsf{certifiedKill}(M_t, M_t, L)$ at

$$t \in \{1, 2, 3, 4, 5, 7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32, 37, 41, 43, 47, 49, 53, 59, 61, 64\};$$

- a finite joint certificate refuting cone flatness at a tabulated family of vertices.

The four validation classes are formalised, in order, by the small-window certificate family, periods through sixteen, 28 imported diagonal certificates through scale 64, and cone-cell certificate table.

For example, at $(h, N, L) = (1, 12, 16)$,

$$\Delta_{1,12,16} = -143140, \qquad \Delta_{1,12,16} \bmod 2^{16} = 53468,$$

while the omitted-tail allowance is $12 + 1 + 16 + 2 = 31$. Hence

$$31 < 53468 < 2^{16} - 31,$$

which is exactly $\mathsf{certifiedKill}(1, 12, 16)$ and therefore proves $R_{13} - R_{12} \notin \mathbb{Z}$. Every row above has this exact status: it proves a particular non-integral tail difference, not an unbounded supply.

5.4 What remains open

By Theorem 5.4, the unresolved statement is precisely

$$\forall h \geq 1 \ \forall N_0 \geq 0 \ \exists N \geq N_0 \ \exists L \geq 1, \qquad \mathsf{certifiedKill}(h, N, L). \tag{5.1}$$

The diagonal hypothesis in Corollary 5.5 is a sufficient one-parameter strengthening of this statement. The verified scales in Example 5.7 form a finite set and do not establish either quantifier pattern. No theorem in the formal source proves (5.1); consequently no irrationality claim for S is made here.

Appendix C records exact reformulations of this open obligation, finite projections and denominator formulae, obstructions to several stronger local criteria, and closed strategy branches. They sharpen the target but do not supply its missing unbounded quantifier.

6 Mathematical architecture of the checked arguments

This section is interpretive: it introduces no additional mathematical claim. Its purpose is to identify which parts of the checked arguments are finite and complete, which parts require a uniform theorem, and why the chosen coordinates expose that distinction.

Finite arithmetic versus uniform supply. The arguments repeatedly separate a finite arithmetic core from a statement that must hold uniformly. In the full-support Erdős–Borwein theorem, the finite core consists of a Bertrand–CRT choice, divisor-pair averaging, and explicit parameter inequalities. For #249 the separation is sharper: eventual binary periodicity predicts an integral tail difference; finite modular arithmetic can refute that prediction at fixed parameters; and Proposition 5.6 proves that the finite certificate loses no information there. The missing ingredient is therefore not a larger finite verification, but a theorem producing such witnesses without bound.

Tail coordinates and completeness. The choice of tail coordinates is essential. Rather than reason about an informal repeating digit string, one uses the scaled tails R_N and the fact that rationality forces suitable period differences to be integers. Certificate completeness then separates two logically distinct tasks: finite arithmetic decides non-integrality at a fixed pair (h, N), while Theorem 5.4 requires those pairs beyond every threshold.

Equivalent representations. Three further changes of coordinates serve different mathematical purposes. The Mersenne–Lambert transform places S next to the Erdős–Borwein constant and exposes the nonnegative primitive-conductor weight. The Möbius-square identity turns the same number into a signed rapidly convergent series. The coprime-pair identity turns it into a probability. Formalising all three and proving that the denominator exclusion transfers between them prevents an informal change of representation from silently changing the claim. Reducing the random pair by its gcd then yields the exact Stern–Brocot cylinder law of Theorem D.8 and the Fibonacci–continuant estimates of Theorem D.9. These are unconditional structure inside the probability representation, not a substitute for the certificate supply.

Necessary conditions on a rational support. The mathematical assertions in this part of the architecture are exactly those of Theorem 4.4, with their proofs and auxiliary coordinates in Appendix B. Their role is diagnostic. The zero-window bound in Theorem 4.4(ii) is obtained by transporting the fixed-power divisor estimate $\tau(n)^k \le (k^{2^k})^k n$ through the binary tail and the exact carry recurrence. Part (i) shows that an infinite rational support cannot be represented by a bounded carry alphabet. Part (iii) relates the odd denominator to reciprocal mass, retaining divergence as an explicit alternative. None of these necessary conditions is a contradiction; they describe the infinite Boolean carry path that a future proof would have to exclude.

General proof template and residual. The same architecture is useful beyond the two named problems: choose a representation in which rationality imposes a rigid tail law, isolate a finite predicate that soundly refutes that law, prove the predicate complete at fixed parameters, and state the remaining uniformity theorem separately. The last step matters as much as the first three. It prevents a large verified finite range from being mistaken for the infinite theorem. Thus the exact consequence of this section is organisational rather than mathematical: fixed-parameter completeness has been separated from the unbounded supply required for #249 and from the two unresolved #257 directions.

7 Formal verification and reproducibility

This section records proof authority and reproducibility; it does not add a mathematical conclusion to either theorem spine.

Proof authority and source identity. The Lean 4 proof assistant [19] and Mathlib library [20] check the named results. The source, exact informal-to-formal links, and build instructions are available in the public repository. This paper is tied to an archived formal-source revision; every blue mathematical caption links directly to the corresponding checked theorem there. The repository pins the theorem prover and library versions needed to reproduce the build.

Computational and expository boundary. Every declaration is elaborated and every proof term is checked by the Lean kernel. Finite computations use kernel-checked decision procedures rather than an external evaluator, and the source contains no proof placeholders or added axioms. The manuscript is exposition rather than proof authority; Appendix E explains how each displayed result is connected to its exact formal statement.

Section consequence. The linked declarations and their checked proof terms certify the stated proved, conditional, finite, and equivalence results at the archived source revision. That verification does not change the mathematical status of the unresolved statements: the open problems collected in Section 8 remain open.

8 Exact frontiers and further questions

The conclusions divide into one open #249 obligation and a two-way #257 frontier. For Erdős #249, Theorem 5.1 excludes a large finite denominator range and Theorem 5.4 gives an exact certificate characterisation; the missing assertion is the unbounded certificate supply (5.1). For Erdős #257, Theorem 4.1 settles full support and Theorem 4.2 settles several structured infinite supports, but the universal statement remains open. A counterexample would instead be a rational member of the achievement set $\mathcal{A}$, with $1/2$ the distinguished target analysed in Section 4. The achievement-set topology and the Stern–Brocot cylinder law are exact structural results on the two sides of this frontier; neither supplies the missing universal or cofinal assertion.

8.1 Two concrete scalar criteria

Two scalar conditions give concrete sufficient routes toward the two open directions. Both are exact arithmetic statements; neither is proved at the unbounded range needed for its final conclusion.

For #249, let m_t be the canonical adjacent-suffix depth and let d_t be the corresponding residue in $[0, 2^{m_t})$. Define

$$\sigma_t = \min\bigl(d_t - 2^{m_t-5}, (2^{m_t} - 2^{m_t-5}) - d_t\bigr) \in \mathbb{Z}.$$

Proposition 8.1 (canonical strict-jump slack)**.** *The canonical residue lies in the central band $[2^{m_t-5}, 2^{m_t} - 2^{m_t-5}]$ if and only if $\sigma_t \geq 0$. If nonnegative slack occurs at arbitrarily large strict jumps $M_t < M_{t+1}$, then S is irrational.*

Formalised: canonical central-band equivalence, strict-jump slack equivalence, and cofinal-slack irrationality criterion. The proposition is an equivalence followed by a sufficient criterion. It does not prove that σ_t is nonnegative cofinally.

For #257, write r_n for the exact greedy residual of the target $1/2$ and put

$$u_n = 4^n(2r_n - 2^{-n}).$$

The second-channel condition is the rational inequality

$$\frac{1}{6} + \frac{37}{56}2^{-n} \leq \left|u_n - \frac{1}{3}\right|.$$

Proposition 8.2 (rational second-channel reduction)**.** *The second-channel condition is decidable over $\mathbb{Q}$ and holds for $1 \leq n \leq 6$. If it holds for every $n \geq 7$, then $1/2$ belongs to the Mersenne achievement set $\mathcal{A}$.*

The finite check and the unbounded-range implication are formalised as finite second-channel verification and the from-seven membership criterion. The membership conclusion is itself exact:

$$1/2 \in \mathcal{A} \quad \Longleftrightarrow \quad \text{the canonical greedy orbit omits infinitely many exponents.}$$

This is formalised at half-membership–infinite-skip equivalence. The forward implication uses Theorem 4.1; the reverse implication uses the absorbing fatal-state criterion. Thus the second-channel hypothesis is one sufficient route to the infinite-skip condition. It does not prove that condition, and membership alone is not the universal #257 theorem.

8.2 The unresolved statements

Problem 8.3 (Erdős #249)**.** Prove that

$$S = \sum_{n \geq 1} \frac{\varphi(n)}{2^n}$$

is irrational.

By Theorem 5.4, the preceding problem is equivalent to the following exact quantifier statement.

Problem 8.4 (unbounded certificate supply)**.** Prove (5.1). The lcm-diagonal supply in Corollary 5.5 is sufficient, but the finite scales in Example 5.7 do not establish it.

Problem 8.5 (universal Erdős #257)**.** Prove that

$$\sum_{n \in A} \frac{1}{2^n - 1}$$

is irrational for every infinite set $A \subseteq \mathbb{N}$, rather than only for the support families in Theorem 4.2.

Problem 8.6 (the half-value counterexample route)**.** Decide whether $1/2 \in \mathcal{A}$. By Theorem 4.6, this is equivalent to the canonical greedy orbit omitting infinitely many exponents. A positive answer would give an infinite support of rational value $1/2$ and refute the universal #257 statement; a negative answer would close this distinguished counterexample route only.

The scalar criteria above isolate two concrete subproblems: cofinal nonnegative slack would settle #249, while the second-channel inequality would settle the half-value membership question. Neither assertion follows from its checked finite range.

Rationality on the #257 side would also impose two complementary forms of rigidity: divisor-coverage zero windows have length $o(\log N)$, whereas an infinite support with rational series value forces the associated positive carry states to be unbounded. These constraints do not yet exclude a rational infinite-support value.

Appendix D collects further weighted-Lambert, Möbius-square, and finite-algebraic identities. They provide additional coordinates and obstructions, but no further closure of the three statements above. Across both problems the missing step has the same logical form: a finite obstruction is exact at fixed parameters, while the theorem requires that obstruction uniformly or without bound.

Statements and declarations

Artefact and data availability. The archived formal-source revision contains the Lean sources, pinned toolchain, library manifest, and generated certificate data used in the verification. Repository metadata and this manuscript provide navigation rather than proof authority; Section 7 gives the verification route.

Use of AI assistance. Large-language-model assistants were used during development for proof drafting, refactoring, and prose editing. Every registered formal claim is linked to a declaration in the archived source revision. Lean checks each proof term against the pinned library; the project contains no proof placeholders or project-defined axioms.

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A Technical reductions for the half-value branch

This appendix records the local arithmetic used to reach Theorems 4.6 and 4.7. Its statements fall into four classes: exact coordinate identities, conditional compactness criteria, bounded finite certificates, and counterexamples to tempting local inductions. None changes the two open conclusions at the end of Section 4. The appendix is included so that the hypotheses in the main reduction can be audited without turning the main theorem section into a chronology of the formal development.

The order is by mathematical use. Appendix A.1 localizes one possible unsafe skip; Appendix A.2 records conditional structured-support criteria; Appendix A.3 collects the finite-window and compactness implications; Appendix A.4 develops the equivalent final-skip coordinates; and Appendix A.5 records verified finite instances, exact coordinate translations, counterexamples, and closed strategy branches. This is a dependency map, not a second route to a conclusion.

Throughout this appendix, a *seam* is an adjacent pair of finite greedy words at a branch boundary, a *rewind* is the corresponding inverse finite recurrence, and the *transition carry* is the integer coordinate appearing in the exact residual identity. These terms denote mathematical objects, not stages of a proof search.

A.1 A band formula for the final skip between two takes

One local configuration admits a simple exact description in full generality. Suppose a take at rank b is followed by the next take at $c > b + 1$. Put $q = 2^b - 1$, $m = 2^{c-1}$, and write the residual immediately before the first take as $1/R$. When $0 < R < q$, the final intervening skip is dyadically unsafe if and only if

$$\frac{q(m-1)}{q+m-1} < R < \frac{qm}{q+m}.$$

The interval has exact width

$$\frac{q^2}{(q+m)(q+m-1)}.$$

Formalised: the general unsafe-band equivalence and the general band-width formula.

For a single intervening skip, $c = b + 2$ and $m = 2q + 2$. The band becomes

$$\frac{q(2q+1)}{3q+1} < R < \frac{2q(q+1)}{3q+2},$$

with width $q^2/((3q + 1)(3q + 2)) < 1/9$, entirely between $2q/3$ and $2q/3 + 2/9$. No integral reciprocal lies in this window. In the corresponding positive odd integer formulation, with residual $p/(2D)$, membership in the cleared band forces $p \geq 7$. Formalised: the one-skip band equivalence, the one-skip width formula, the width bound, the integral-reciprocal exclusion, and the odd-coordinate lower bound.

These are localizers, not orbit-level eliminations. They do not show that the greedy orbit of $1/2$ avoids any such band, and they do not identify the two remaining exceptional transition coordinates $-2, -1$ with dyadically unsafe skips.

A.2 Conditional criteria for structured supports

A.2.1 A conditional pairwise-coprime dilation criterion

Let A be a finite-core orthogonal-petal bouquet: outside a finite frame, its elements have the form $c_i p_i$, where the cores c_i divide one fixed integer and the nontrivial petals p_i are pairwise coprime to that frame and to one another, with summable reciprocal mass. The formal construction also contains an explicit infinite example with alternating cores 2 and 3 and squared prime petals; every member of this support has exactly three prime factors with multiplicity.

For such a bouquet, the remaining input is an exact tail-selection predicate: at every positive binary scale it chooses a carry slot with a normalized tail at most 16. Lean proves that this predicate supplies the needed forced-carry certificates and hence irrationality of the base-2 support series. The result is the conditional bouquet irrationality theorem, with the carry bridge at the forced-carry implication. The tail-selection predicate itself is not proved here. Thus this is a conditional reduction for one structured support family, not a further unconditional case of Erdős #257.

A.2.2 Composite dilation defects

For a support divisor count $f_A(x) = \#\{d \in A : d \mid x\}$, multiplication by a composite support element a has an exact correction term. Besides the distinguished divisor a, the new divisors are counted by a finite defect which excludes divisors already present at x. The formal identity is the composite-dilation identity. For a finite-core orthogonal-petal bouquet, the defect at a ray is bounded by the finite exceptional frame plus the divisor count of the petal support: the bouquet defect bound. This records the local correction needed when dilation is not prime. It does not prove a correlation estimate, the sunflower tail selector, or an irrationality statement.

A.2.3 Mixed prime-power layers

The one-prime-power layer has a two-signature extension. Exact p-adic and q-adic layer operators commute, and, for distinct primes with the residual argument coprime to pq, their composite extracts the iterated exact-valuation pullback of the support coefficient. This is formalised by the two-prime layer identity. The singleton support $\{12\}$ gives a checked $(2^2, 3)$ example with value one at residual argument 1: the singleton-support calculation. These are finite coefficient identities. They do not provide a decimation transport, a bounded-Ω conclusion, or an irrationality theorem.

A.2.4 Half-trapping return carries

If rational-linear channels vanish on every relation killed by one scalar evaluation, their evaluation matrix has rank at most one; every nontrivial square minor therefore vanishes. The all-rank determinant statement is the rank-one determinant criterion. For two reverse carries with a common coefficient word, a $1/0$ seam, and a common following overlap, the terminal difference is a power of two times an odd integer. A shared nonnegative Archimedean bound then gives the matching dyadic spacing inequality, formalised at the dyadic spacing inequality. The argument does not produce reverse-carry presentations or an earlier overlapping return; it supplies only the algebraic core of that conditional criterion.

A.2.5 Half-carry reachability

There is also a finite-prefix criterion for the target $1/2$. At an unresolved even seam, the scalar transition reaches every integer terminal carry except the single value $2\delta - c$; this is the exact equivalence the even-seam reachability equivalence. A canonical description of such a seam at every sufficiently large even depth is an unproved hypothesis. Under that hypothesis, the finite admissible words are cofinal, by the cofinal-admissibility implication, and compactness then produces an infinite support with sum exactly $1/2$: the half-value compactness conclusion. Thus the formal result is a conditional reduction. It does not establish the canonical seam supply and does not prove Erdős #257.

Cofinal strip returns. The all-level strip hypothesis can be weakened for the canonical greedy half carry. Its uncentred, binary-scaled form is antitone. Consequently, if it returns to the square-root strip cofinally, then one sufficiently late return bounds every later scaled carry; the strip envelope divided by 2^N tends to zero. The resulting tempered centred carry gives an infinite greedy support with support-series value $1/2$.

Proposition A.1 (cofinal strip-return criterion)**.** *If the canonical greedy half carry returns cofinally to the stated square-root strip, then* $1/2$ *belongs to the Mersenne achievement set.*

Formalised: scaled-carry decay and the strip-return membership criterion. The cofinal return condition itself is not established. It is weaker than a uniform strip bound and still does not give an unconditional membership theorem.

Terminal-only cofinal approximation. Coherence can be removed entirely from a further version of this reduction. At depth M, take only a normalised finite word whose terminal half carry lies in the square-root strip. Its finite support then has support-series value within

$$\frac{4M+12}{2^M}$$

of $1/2$. If such terminal witnesses occur cofinally, these finite values converge to $1/2$; closedness of the Mersenne achievement set supplies the limit. No control of the earlier rows, and no compatibility between the witnesses, is used.

Proposition A.2 (terminal-only cofinal criterion)**.** *Cofinal normalised terminal witnesses in the square-root strip imply that* $1/2$ *belongs to the Mersenne achievement set, and hence that an infinite support has support-series value* $1/2$*.*

Formalised: the terminal approximation bound, the terminal-strip compactness criterion, and the infinite-support conclusion. The cofinal terminal-witness supply is not proved. In particular, the proposition does not establish membership of $1/2$ unconditionally.

Scaled terminal vanishing. The square-root strip is stronger than the analytic condition used by the terminal-only argument. Let the depths of normalised finite words tend to infinity. If their terminal carries, after division by the corresponding power of 2, tend to zero, then the universal coefficient-tail error also tends to zero. The finite support values therefore converge to $1/2$, and closedness of the achievement set again supplies the limit. The earlier terminal-only strip condition implies this weaker sequential hypothesis.

Proposition A.3 (scaled terminal-vanishing criterion)**.** *A normalised terminal-word sequence with depths tending to infinity and vanishing binary-scaled terminal carries yields an infinite support with support-series value* $1/2$*.*

Formalised: the scaled terminal error bound, the scaled-vanishing compactness criterion, and the scaled-vanishing support conclusion. No such scaled-vanishing sequence is constructed here; the result is a conditional reduction, not an unconditional membership theorem.

A.3 Finite-window reductions and compactness criteria

A.3.1 Finite shadows of the half cylinder

For an admissible finite word, the residual from $1/2$ is given exactly by its terminal half carry minus the future coefficient tail, scaled by a power of two; see the finite-shadow residual identity. At a first seam, avoiding the integer strip is equivalent to full scalar reachability of that strip: the seam-escape equivalence. The rank-three greedy example is raw dyadically safe while its seam has an in-strip hole, so raw safety alone does not imply this escape. Finally, an explicit frozen-margin hypothesis implies $1/2$ belongs to the Mersenne achievement set, by frozen-margin implication. That hypothesis is unproved; this is a conditional reduction, not a proof of Erdős #257.

A.3.2 Fixed-coefficient rewind

For a finite list of common coefficients, the canonical inverse-parent map has an exact closed form with a power-of-two denominator: the rewind closed form. An interval of width at most that denominator contracts either to one ancestor or to two adjacent ancestors. The singleton alternative is equivalent to the explicit dyadic phase inequality, while the other alternative is a seam pair; see the singleton-or-seam dichotomy. Thus width alone does not justify a singleton conclusion. This is a finite rewind lemma, and supplies neither a seam exclusion nor an Erdős #257 conclusion.

A.3.3 Selected half-carry windows

The selected-window construction keeps an explicit admissible word for each terminal carry in a finite protected interval. A supply of such windows at all sufficiently large levels gives cofinal admissibility and hence an infinite support with sum $1/2$: selected-window cofinality and the selected-window half-value conclusion. The unresolved input is the coherent history at each step: either a common next-row divisor cylinder, or the canonical adjacent seam. The theorem states this input exactly; it does not construct the window supply and does not prove Erdős #257.

A.3.4 A realized rewind criterion

For a realized fixed-coefficient history, the dyadic rewind phase is the complementary binary suffix numeral: the suffix-phase identity. Consequently the singleton criterion is exactly the inequality that this suffix numeral covers the rest of the terminal interval, formalised by the suffix-coverage criterion. This converts the phase condition into support-history coordinates, but does not prove the required coverage inequality globally.

A.3.5 Localized protected seams

The compactness argument needs only one admissible word at cofinally many depths. At a localized one-hole seam, one of the two protected carries 3 and 4 survives; therefore a cofinal supply of realized protected seams implies cofinal admissibility by the protected-seam cofinality implication, and hence an infinite support with sum $1/2$, by compactness. The cofinal realization supply is not proved.

A.3.6 Combining the two cofinal half-carry criteria

Assume that at each requested scale there is either a selected protected window or a localized realized seam. Each alternative supplies the one admissible word needed for compactness, so this cofinal disjunction implies an infinite support with sum $1/2$: by window-or-seam cofinality and by the window-or-seam half-value conclusion. The cofinal disjunction remains unproved.

A.3.7 Integer-greedy first-wrap reduction

The first-wrap seam has exact truncated Mersenne weights with a quantitative gap-dominance property. For these weights, a small positive subset-sum defect exists exactly when the descending integer-greedy remainder is small and positive; this is formalised by the integer-greedy remainder equivalence. The remaining arithmetic input is an unbounded lower bound for that deterministic remainder sequence. No such lower bound is proved here.

A.3.8 Exact finite recurrence for the greedy seam

The finite seam-word construction realizes the abstract adjacent cut by concrete Boolean words and identifies its next word with the integer-greedy choice: the seam-word recurrence. It implements a finite recurrence, not the unbounded remainder estimate.

A.4 Fatal gaps and the final-skip reduction

A.4.1 Fatal gaps and eventual right tails

For the concrete integer seam words, eventual right extension is equivalent to nonmembership of $1/2$ in the Mersenne achievement set: the eventual-right nonmembership equivalence. The same condition is equivalent to the existence of a finite fatal half gap, by the fatal-gap equivalence, and also to a final skipped exponent in the real greedy orbit, by the final-skip equivalence. Thus this equivalence gives an exact classification of the nonmembership branch. It neither decides whether $1/2$ belongs to the achievement set nor proves the universal Erdős #257 statement.

This is the nonmembership half of the argument summarized earlier: a last greedy skip is equivalent to eventual right extension. The following paragraph records the first exceptional middle case that can be ruled out.

First final-middle cell. The first exceptional middle cell has a separate obstruction: if the orbit extends right after the last false rank, then the cell with transition carry -3 makes the lazy cofinite tail strictly smaller than $1/2$ while its centred carry becomes negative. This is impossible by the analytic nonnegativity identity, the middle--3 exclusion. It removes that one conditional terminal configuration; it does not decide the right-tail branch or the remaining two middle cells.

The remaining global reduction is now precise. If every genuine middle transition other than that discharged cell has transition carry larger than its complete future divisor-incidence tail, then an eventual right seam is impossible and $1/2$ belongs to the Mersenne achievement set: middle-tail implication. The stronger square-root lower-bound hypothesis is also sufficient, square-root sufficient condition. Neither hypothesis is supplied here.

For finite seam prefixes the remaining tail bound sharpens further: the future divisor-incidence tail is bounded by the cardinality of the prefix. Thus the explicit row-scale inequality $|c| + \mathrm{belowPulse} + 5 < 4\,\mathrm{remainder}$ is sufficient for the same half-membership conclusion, prefix-cardinality bound; it is enough in turn that every late middle remainder is at least its row, row-scale sufficient condition. These are exact conditional reductions, not proofs of the row bound.

Finite upper-reset band certificates. The companion upper branch has a kernel-checked finite certificate: for every upper transition from rows 13 through 30, every subsequent dyadic danger band is avoided. The certificate shares each exact successor remainder across all its band indices, the verified upper-reset certificate. It is a bounded transition check and does not supply the cofinal band-escape hypothesis.

A.4.2 Positive half-membership classification

The complementary branch has an equally exact seam formulation. Membership of $1/2$ in the Mersenne achievement set is equivalent to false successor terminal bits beyond every bound, the unbounded terminal-skip equivalence; equivalently, it is equivalent to a cofinal sequence of such false terminals, the cofinal terminal-skip equivalence. It is also equivalent to a cofinal integer-seam sequence with skipped ranks tending to infinity, the unbounded skipped-rank equivalence. These equivalences isolate the required cofinal skipped-rank condition; they do not establish that condition.

A.4.3 Largest skipped ranks

At a named largest skipped rank d, the lower seam word and its adjacent upper competitor have an exact late-range gap. In division-free form, the identity is

$$3L + \left(3 \cdot 2^{s+1} + 2 \cdot 4^{s-d} + 4\right) = 3U$$

whenever $2s < 3d$; this is formalised at the exact late-gap identity. The upper and middle successor branches create a terminal largest skipped rank, while the right branch preserves an existing one: the terminal largest-skip branch and right-branch persistence. The first-crossing branch hypothesis required to use this identity globally is not proved.

A.4.4 Boundary-pulse normalization

Above a late largest skipped rank, every filled suffix rank has zero row pulse, so the word pulse is carried entirely by the lower prefix: the lower-prefix pulse localization. At the first crossing of the two-thirds boundary there are exactly two cases, $3d = 2s + 1$ and $3d = 2s + 2$, and the skipped rank has pulse respectively two or one: the first-crossing pulse classification. This supplies the incidence normalization for a transition-carry estimate; it does not establish that estimate.

A.4.5 Largest-skip induction

Assume that at the first crossing a late largest skipped rank either remains late at the next row or forces an upper-or-middle successor branch. This hypothesis propagates a late largest skipped rank from row 14 onward, largest-skip propagation step, and hence yields cofinal skipped ranks and membership of $1/2$ in the Mersenne achievement set, largest-skip membership criterion. The first-crossing hypothesis itself remains unproved.

A.4.6 The fixed-tail survival criterion

A skipped rank is final exactly when its post-decision remainder lies strictly above the complete remaining Mersenne tail: the final-skip criterion. Consequently membership of $1/2$ is equivalent to the pointwise assertion that every actual skipped rank has remainder at most its full future tail, the half-membership survival equivalence. This is an exact local reformulation; the tail-survival inequality is not proved.

A.4.7 The carry identity at a non-right transition

For a finite support aligned with a non-right transition, the residual from $1/2$ is exactly the integer transition carry minus the complete future divisor-incidence tail, scaled by $2^{-(2d+2)}$:

transition-carry residual identity. Thus the carry exceeds that tail exactly when the support series lies below $1/2$, and a negative carry places it above $1/2$; positive-carry criterion and negative-carry criterion. An upper carry at most -8 even gives a full Mersenne-gap margin, full-gap carry criterion. These are local criteria, not a proof that the required carries occur.

A.4.8 Quarter-band endpoint cells

For the adjacent seam recurrence, failure of the desired next-row upper bound on the middle branch is exactly one multiple-of-four pulse cell, the middle-branch pulse-cell equivalence. On the terminal-reset branch, the same failure is exactly a constrained $3 \cdot 2^{s-1} + k$ pulse cell, the reset-branch pulse-cell equivalence. These are arithmetic normal forms for the exceptional cells; no avoidance statement for the concrete seam orbit is proved.

A.4.9 Reset-deficit escape

At a late largest skipped rank, a right successor branch pins the integer greedy remainder between an explicit quarter threshold and carry threshold: the right-branch remainder window. Along such a branch the remainder satisfies an exact affine recurrence, the right-branch affine recurrence. Moreover, a doubled-scale deficit forces an upper-or-middle branch within the corresponding number of rows, the deficit escape bound. The remaining input is the deterministic anti-concentration needed to exclude the pinned right-branch window globally.

A.4.10 Alignment of seam and transition carries

At a first-wrap seam, the transition carry of the left boundary support is four times the shifted seam hole, less the explicit paired incidence pulse at the next two rows: seam–carry alignment. This is an exact coordinate change between the seam and transition coordinates; it does not supply a sign or size estimate for that carry.

A.5 Finite computations, exact identities, and counterexamples

A.5.1 A verified initial selected window

The selected half-carry construction has a checked depth-18 base: twelve Boolean words cover every terminal carry in the strip $[1, 12]$, with all intermediate carries in the corresponding half strip. The reflected table is formalised at the depth-18 table verification; the resulting selected window is defined at the depth-18 selected window. All twelve words share the same restriction at depth 13, giving the first next-row divisor-agreement certificate, the depth-18 divisor-agreement certificate. This finite base does not provide selected windows at unbounded depths.

A.5.2 Rewind-to-seam equivalence

For a selected window with a rewind seam, a two-value coefficient profile on the adjacent base ancestors induces the live next-row profile, the next-row rewind profile. Under the stated strip and buffer inequalities, that profile realizes a protected one-hole seam at the next row, the protected-seam realization. The base-ancestor coefficient drop is an explicit input, not a theorem here.

A.5.3 Selected suffix cylinders

Put

$$H_N = 2\lfloor\sqrt{N}\rfloor + 4.$$

For a Boolean word $a = (a_0, \dots, a_N)$, let

$$f_a(n) = \#\{d \le N : d \mid n,\ a_d = 1\}, \qquad K_0(a) = 1, \qquad K_{r+1}(a) = 2K_r(a) - f_a(r+2).$$

Definition A.4 (selected window and suffix cylinder)**.** A *selected window* of depth N and radius R is a family $(a^{(k)})_{1 \le k \le R}$ of Boolean words with $a_0^{(k)} = a_1^{(k)} = 0$ such that

$$1 \le K_{n-1}(a^{(k)}) \le H_n \quad (1 \le n \le N), \qquad K_{N-1}(a^{(k)}) = k.$$

For $M \le N$, define the suffix numeral

$$\nu_{M,N}(a) = \sum_{j=M+1}^{N} a_j \, 2^{N-j}.$$

The window has a *suffix cylinder at cutoff M with endpoint E* if all $a^{(k)}$ have the same prefix through M and

$$\nu_{M,N}(a^{(k)}) + k = E \qquad (1 \le k \le R).$$

A *profiled gap stage* additionally permits one missing carry interval $[g_-, g_+] \subseteq [1, H_N]$: a lower suffix-cylinder family represents $1, \dots, g_- - 1$, an upper family represents $g_+ + 1, \dots, H_N$, and the two families retain fixed consecutive prefixes through the cutoff.

Proposition A.5 (conditional suffix-cylinder feedback)**.** *The following statements hold.*

(i) *The verified depth-*18 *selected window has* $R = 12$ *and a suffix cylinder at cutoff* $M = 13$ *with endpoint* $E = 17$.

(ii) *Whenever the canonical one-row successor hypotheses hold and every selected word has the same next coefficient C, the successor remains a suffix cylinder and its endpoint is*

$$E' = 2E - C.$$

At a feedback row $N + 1 = 2(M + 1)$, either the full cylinder advances or the exceptional carry values form a profiled gap stage; the alternative includes the protected-seam case.

(iii) *Suppose a profiled gap stage has lower and upper next coefficients C_- and C_+, respectively. If*

$$\begin{gathered} 1 \le 2g_- - C_- - 1, \qquad C_+ \le 2g_+, \\ 2g_- - C_- - 1 \le 2g_+ - C_+ \le H_{N+1}, \\ H_{N+1} + C_+ \le 2H_N, \end{gathered}$$

then the stage advances one row with exact child gap

$$[g'_-, g'_+] = [\,2g_- - C_- - 1,\ 2g_+ - C_+\,].$$

At a cutoff boundary $N = 2M + 1$, $M \ge 4$, the lower and upper prefixes acquire the forced bits 0 *and* 1*; their consecutiveness is preserved. Under the displayed child-gap conditions at the next row, this promoted stage therefore advances with coefficients determined by its retained prefixes.*

(iv) *If a two-row update swallows a protected strip of bound B, and the lower coefficient pulse is at most the upper pulse plus* 2*, then*

$$B \le 4(g_+ - g_- + 1) + 2. \tag{A.1}$$

This is a necessary quarter-band condition, not a contradiction.

(v) *If full suffix-cylinder stages exist at arbitrarily large depths, then there is an infinite set* $A \subseteq \mathbb{N}$ *such that*

$$\sum_{n\in A} \frac{1}{2^n - 1} = \frac{1}{2}.$$

Part (i) is the depth-18 cylinder. Part (ii) is the endpoint recurrence, feedback dichotomy, and two-family gap alternative. Part (iii) is the two-coefficient gap recurrence and promote-and-advance theorem. Part (iv) is the swallow bound. Part (v) is the cofinal-cylinder criterion.

Status and prior-art boundary. The proposition is a conditional propagation theorem: it does not supply the coefficient inequalities or the cofinal family in part (v). The strict-tail achievement-set literature supplies the surrounding Cantor-set geometry [14], not this finite carry recurrence; the comparison record makes no novelty claim for the recurrence itself.

A.5.4 Verified finite suffix-cylinder stages

The finite record has three milestones.

(1) *Depth* 27. Before the first half-divisor feedback row, a full selected window exists at every depth from 18 through 27, and the depth-27 endpoint covers the whole protected strip: stages through depth 27 and depth-27 strip coverage.

(2) *Depth* 29. The threshold-stage continuation still covers the strip and has a common suffix cylinder at every cutoff from 14 through 25, while cutoff 26 is explicitly not common: depth-29 strip coverage and cutoff-26 obstruction.

(3) *Depth* 52. An independent exact run reaches depth 51 with endpoint 51,327,745, beyond the next feedback head threshold, and the promoted successor at depth 52 is checked: depth-51 threshold margin and depth-52 terminal-strip witness.

These are finite milestones, not an unbounded induction; they do not provide the arbitrarily large stages required by Proposition A.5(v).

A.5.5 Half-divisor unit drop

At the doubled feedback row, changing only the terminal half-divisor bit from false to true increases the support coefficient by exactly one: the terminal-bit unit-drop identity. The same holds for any explicitly identified adjacent boundary pair, the boundary-pair unit-drop identity. The inverse-parent recurrence transfers such a pair to the next protected seam while retaining the exact unit-drop witness, the rewind unit-drop transfer and the protected-seam transfer. This is the local coefficient input for the rewind-seam profile, not a global construction of such boundary pairs.

A.5.6 A finite rewind-boundary obstruction

The scalar one-row rewind data alone still does not supply that missing boundary predicate. Lean gives a concrete depth-26 selected two-word window whose depth-27 explicit step has rewind history [1], is a width-two seam pair, and satisfies the doubled-row identity, but has no depth-13 half-divisor boundary-pair witness: the depth-27 boundary-pair counterexample. Thus the protected-window buffer in the compactness criterion is a genuine hypothesis, not a consequence of the scalar seam facts alone.

A.5.7 Explicit carry coordinates at the final non-right transition

For every adjacent cut at rank $s \geq 5$, the transition carry of the terminal-augmented below support equals the concrete middle coordinate:

$$4\,\mathrm{remainder} - \mathrm{belowPulse} - 4,$$

and identifies the terminal-augmented above support with the negative upper coordinate

$$-\big(4\,\mathrm{overshoot} + \mathrm{abovePulse} + 4\big).$$

The generic row-word identity is row-word transition identity, with the below and above specializations at middle-coordinate identity and upper-coordinate identity. These identities make the remaining quantitative task exact: they do not prove exponential lower bounds, sign persistence, or a universal escape mechanism.

A.5.8 Strategy boundary: Campbell shift synchronization

A proposed route combines a finite Mersenne shift from a putative last greedy skip with Campbell's quarter-exponent prime progression. The exact parameter requirements cannot be synchronized: if the combined progression modulus is $d \geq 2$, the phase factor is at most d, the fatal-window prime cap is at most four times that phase, and Campbell's applicability requires d^4 below the cap, then the inequalities are inconsistent. Formalised: the period-freeze incompatibility and the parameter incompatibility.

The associated shifted-zero condition is not an independent hypothesis. It is exactly equivalent both to infinitely many actual greedy skips and to $1/2 \in \mathcal{A}$: the shifted-zero/skip equivalence and the shifted-zero/half-membership equivalence. This closes one synchronization strategy and one reformulation; it does not advance either endpoint beyond the already stated half-achievement problem.

Appendix consequence. The compactness and final-skip subsections give exact implications: a cofinal window, seam, or terminal approximation would produce $1/2 \in \mathcal{A}$, whereas an eventual right tail is exactly nonmembership. The other subsections localize possible inputs, verify bounded instances, and close stronger but invalid local inferences. None produces the required cofinal input, so the residual remains precisely the two obligations in Theorem 4.7.

B Auxiliary rationality criteria from binary carry systems

Five further theorem families provide additional criteria. They prove no new irrationality. Instead they pin down, in machine-checked form, what a rational value of the series in question would have to look like: as an integer carry orbit, as a Möbius-inverted support coordinate, as a residue-wrap cycle modulo the odd part of the denominator, and as a point of a greedy subset-sum geometry. Everything in this section is an equivalence or an unconditional constraint on the rational side; nothing excludes a rational value of either open series, and each family states that boundary explicitly. The prior-art boundary is theorem-family-specific. The integral carry recurrence has a direct antecedent in Wang–Grau Ribas [15]; the strict-tail geometry is recorded by Kovač–Tao [14]; Möbius inversion, repetend algebra, and divisor averaging are classical (see, for example, Apostol [6]). We make no claim of mathematical priority for the converse/rigidity, certificate-normal-form, or coupled reciprocal-mass/collision statements collected here.

The reading order is: a generic tail-orbit rationality normal form in Appendix B.1; its Boolean–Möbius specialization in Appendix B.2; divisor-coverage constraints in Appendix B.3; denominator-wrap, reciprocal mass, and unbounded-state constraints in Appendix B.4; and the exact greedy geometry with one-sided finite death certificates in Appendix B.5.

B.1 Tail-orbit rigidity: a rationality criterion

For a coefficient sequence $c : \mathbb{N} \to \mathbb{N}$ with $c(n) \le n$, write

$$X_c = \sum_{n\ge1} \frac{c(n)}{2^n}, \qquad T_c(N) = \sum_{j\ge1} \frac{c(N+j)}{2^j},$$

the series and its scaled tail after N binary digits. Call an integer sequence u a *tempered orbit* for c with multiplier $v \ge 1$ if

$$u(N+1) = 2\,u(N) - v\,c(N+1) \ \text{ for all } N, \qquad \text{and} \qquad u(N)/2^N \to 0.$$

The recurrence is exact binary long division against the coefficient stream; the boundary condition forbids the homogeneous solution. Wang and Grau Ribas use the integral carry recurrence forced by rationality for the weighted binary special case $c(n) = nd_n$ [15]. The theorem below extends that forward mechanism to arbitrary nonnegative $c(n) \le n$; its explicit converse and subexponential-orbit rigidity are the local additions, with no priority claim attached. Wang–Grau Ribas's positive-density theorem and its Erdős #260 corollary are not used or formalised here.

Theorem B.1 (tail-orbit rigidity)**.** *Let $c(n) \le n$ for all n. Every tempered orbit is the scaled tail, $u(N) = v\,T_c(N)$ for all N; and X_c is rational if and only if a tempered orbit exists for some multiplier $v \ge 1$.*

Formalised: the scaled-tail orbit identity (rigidity) and the rationality–orbit equivalence (the criterion), restated in Mathlib's irrationality vocabulary at the non-irrationality–orbit equivalence. The engine is one observation: a real orbit with $d(N+1) = 2\,d(N)$ and $d(N) = o(2^N)$ vanishes identically (zero-tail rigidity).

The tempered boundary is essential: the homogeneous parasite $N \mapsto 2^N$ can be added to any orbit without disturbing the recurrence, so positivity of an orbit certifies nothing by itself, and positivity is nowhere advertised as an equivalent criterion. The linear-growth hypothesis covers both constants introduced in Section 1: $\varphi(n) \le n$ for the #249 coefficients, and $f_A(n) \le \tau(n) \le n$ for the #257 support coefficients $f_A(n) = \#\{a \in A : a \mid n\}$. The results below instantiate the criterion on supports; the totient instantiation is not taken here.

Proposition B.2 (finite-level carry anti-compression)**.** *Suppose a tempered integral carry orbit for the totient sequence has positive multiplier. If the canonical level-e totient kernel family is linearly independent, then the span of its canonical carry sections has dimension at least $2^e - 1$. Consequently, an all-level independence hypothesis together with rationality of the totient coefficient series would force one tempered carry orbit with unbounded finite-level section ranks.*

Formalised: the carry-kernel rank lower bound and the unbounded-rank consequence of rationality. The all-level independence hypothesis is not proved here. Thus the proposition records an exact conditional anti-compression consequence, not an irrationality criterion for S.

B.2 Boolean–Möbius carry coordinates (the #257 direction)

At base 2 the support series of Section 4 is itself a binary coefficient series, $\sum_{n\in A} 1/(2^n - 1) = X_{f_A}$ (the support-series/binary-series identity). Two exact coordinate changes then turn a hypothetical rational value into a support-free object. The divisor transform and Möbius inversion below are classical Lambert-series tools [6, 7, 8]; the contribution here is their Lean-checked composition with the tempered-orbit criterion.

Proposition B.3 (Möbius-sign support no-go)**.** *Let*

$$A_- = \{d \ge 2 : \mu(d) = -1\}, \qquad A_+ = \{d \ge 2 : \mu(d) = 1\}.$$

Then
$$\sum_{d\in A_-}\frac{1}{2^d-1}=\frac{1}{2}+\sum_{d\in A_+}\frac{1}{2^d-1}\geq\frac{1}{2}+\frac{1}{63}>\frac{1}{2}.$$
In particular, the signed identity $L(\mu)=1/2$ cannot be converted into a Boolean support of value $1/2$ merely by selecting the negative Möbius indices.

Formalised: the negative-Möbius decomposition and the strict lower bound. The term $d=6$ supplies the displayed $1/63$. This closes one sign-truncation route only; it does not exclude another infinite Boolean support with value $1/2$.

Proposition B.4 (Boolean Möbius inversion, both directions)**.** *On positive integers, $f_A=\mathbf{1}_A*\zeta$ and $\mu*f_A=\mathbf{1}_A$, where $\mathbf{1}_A$ is the indicator of A. Conversely, every integer-valued arithmetic function f whose Möbius transform is Boolean, $(\mu*f)(n)\in\{0,1\}$ for all $n\geq 1$, satisfies $f=f_B$ for the support $B=\{n\geq 1:(\mu*f)(n)=1\}$ selected by that transform.*

Formalised: the positive-support zeta identity, the Möbius inversion identity, and, with no search or finiteness hypothesis, the converse the Boolean reconstruction theorem.

On the carry side, Theorem B.1 applies at $c=f_A$: the support series is rational exactly when a tempered carry orbit for f_A exists (the rationality–carry equivalence); a displayed fraction p/q corresponds to an orbit U with multiplier q and initial value $U(0)=p$ (the support-fraction carry equivalence); and exact division recovers the coefficient from the orbit, $\big(2U(N)-U(N+1)\big)/q=f_A(N+1)$ (carry-quotient recovery). The elementary divisor-pair bound $\tau(n)\leq 2\lfloor\sqrt{n}\rfloor$ confines the tails to a square-root strip, $T_{f_A}(N)\leq 2\sqrt{N}+4$. For a nonempty support the orbit states are positive and satisfy $U(N)\leq q\,(2\sqrt{N}+4)$ (the carry upper bound); and the strip is strong enough to re-derive the tempered boundary, so the analytic side condition can be traded for one inequality. The two coordinate changes compose.

Theorem B.5 (certificate-level equivalence)**.** *Fix an integer p and an integer $q\geq 1$. There is a nonempty support A of positive integers with $\sum_{n\in A}1/(2^n-1)=p/q$ if and only if there is an integer sequence U with $U(0)=p$ such that: every $U(N)$ is positive; $U(N)\leq q\,(2\sqrt{N}+4)$; q divides $2U(N)-U(N+1)$ for every N; and the Möbius transform of the quotient sequence $\big(2U(N)-U(N+1)\big)/q$ is Boolean. The support is not guessed: it is reconstructed from the certificate by Proposition B.4.*

Formalised as the normalized support-fraction certificate equivalence.

Example B.6 (the support $\{2,3\}$)**.** The support $\{2,3\}$ has value $\frac{1}{3}+\frac{1}{7}=\frac{10}{21}$ (the two-point series value), its orbit is the pure period-six cycle $10,20,19,17,13,26$ with multiplier 21 (the tempered two-point orbit), and the Möbius transform of the carry quotient recovers exactly $\{2,3\}$ (support recovery).

These are coordinates, not by themselves a strategy: nothing here excludes infinite Boolean carry paths, and no finite search is promoted to a completeness argument.

B.3 Sublogarithmic divisor coverage

The coefficient $f_A(n)$ counts the elements of A dividing n. Say that a zero window of length h starts after $c+N$ when $f_A(c+N+j+1)=0$ for every $0\leq j<h$. Thus no integer in that interval is divisible by an element of the support.

Theorem B.7 (sublogarithmic zero windows)**.** *Let A contain a positive integer and suppose*
$$\sum_{a\in A}\frac{1}{2^a-1}=\frac{p}{2^c v},\qquad p\in\mathbb{Z},\quad c\in\mathbb{N},\quad v\geq 1.$$

For every $\varepsilon > 0$ there is a constant $B \geq 0$, depending only on ε, c, and v, such that every zero window after $c + N$ satisfies

$$h \leq \varepsilon \log_2(N+1) + B, \qquad \log_2 x := \frac{\log x}{\log 2}.$$

The bound is uniform in A, p, N, and h.

Formalised: sublogarithmic zero-window bound.

The proof first establishes, for each integer $k \geq 2$, the explicit divisor estimate

$$\tau(n)^k \leq \left(k^{2^k}\right)^k n.$$

It transports the resulting $n^{1/k}$ bound through the binary tail and then uses the exact carry recurrence across a zero window. Raising the resulting inequality to the kth power absorbs the occurrence of h on the right and leaves a coefficient $1/(k-1)$ in front of the binary logarithm. Choosing k after ε gives the theorem. This controls gaps in divisor coverage; it does not assert a subpower bound for the carry state itself.

B.4 Residue wraps, reciprocal mass, and unbounded carries

Suppose now the support series equals $p/(2^c v)$ with v odd and A containing a positive element. Rationality makes the scaled tails integers: $u(N) = v\, T_{f_A}(c+N)$ is a positive integer, satisfies the carry recurrence $u(N+1) + v\, f_A(c+N+1) = 2\, u(N)$, and reduces modulo v to the doubling residues $r_N = p\, 2^N \bmod v$ (shifted natural-state existence). Doubling a least residue either stays under v or wraps past it exactly once, so each step emits a wrap digit $k_N = \lfloor 2r_N/v \rfloor \in \{0,1\}$, and over any cycle length h with $2^h \equiv 1 \pmod{v}$ the repetend identity holds:

$$\sum_{N<h} r_N \;=\; v \cdot w, \qquad w = \sum_{N<h} k_N,$$

with $w \geq 1$ when p is coprime to $v > 1$. The formal development records the displayed equality as the wrap-count identity and its non-vanishing as the positive-wrap theorem.

Proposition B.8 (one-wrap classification)**.** *Let $v > 1$ be odd, p coprime to v, and $h \geq 1$ with $2^h \equiv 1 \pmod{v}$. If the cycle of p wraps exactly once, then $v = 2^h - 1$ and the starting residue is a power of two: $r_0 = 2^a$ for some $a < h$.*

The one-wrap cycles are thus exactly the Mersenne repetends $2^a/(2^h - 1)$. Formalised: one-wrap classification, from the generic closed-cycle form one-wrap cycle classification, and instantiated at the true order $h = \operatorname{ord}_v(2)$ (odd-order one-wrap classification). The classification is algebraic; a finite check of 446 coprime starting residues across twelve modulus/order rows is retained as kernel-checked validation, not as the proof (446-start finite validation).

The analytic bridge is Cesàro averaging. Write $\rho(A) = \sum_{a \in A} 1/a$ for the reciprocal mass. When the reciprocal family is summable, the mean of $f_A(1), \ldots, f_A(N)$ converges to $\rho(A)$ — the density of the multiples of a, summed over the support — and the mean of the scaled tails has the same limit (divisor-mean limit, coefficient-tail mean limit). The divergent case is carried as an explicit disjunction, not through a default value assigned to a divergent formal sum. Each tail state splits exactly as $u(N) = v\, e_N + r_N$ with integral excess $e_N = \lfloor u(N)/v \rfloor \geq 0$.

Proposition B.9 (order-wrap lower bound, with exact excess mean)**.** *Let A and $p/(2^c v)$ be as above with $v > 1$ odd, and let w be the wrap count of p over one cycle of length $h = \operatorname{ord}_v(2)$. Then $\sum_{a \in A} 1/a$ diverges or $\rho(A) \geq w/h$; if moreover p is coprime to v, then $w \geq 1$, so $\rho(A) \geq 1/\operatorname{ord}_v(2)$. In the summable case the bound is the periodic part of an exact identity: the Cesàro mean of the excess e_N converges automatically, and*

$$\rho(A) \;=\; \frac{w}{h} \;+\; \lim_{N \to \infty} \frac{1}{N} \sum_{n<N} e_n.$$

The divergent-or-bounded conclusion is formalised by the wrap-ratio mass alternative, its coprime specialization by the odd-order mass alternative, and the automatic excess limit in the fraction-facing coordinates above by the shifted excess-mean identity. A rational value thus ties the odd part of its denominator to the sparsity of its support: a summable support with small $\rho(A)$ can only display denominators whose odd part has small multiplicative order of 2.

Corollary B.10 (collision strengthening at common multiples)**.** *In the summable case, let $F \subseteq A$ be finite and let L be a common multiple of F. Then*

$$\rho(A) \;\geq\; \frac{w}{h} \;+\; \frac{\lceil (|F|-1)/2 \rceil}{L}.$$

In particular, an infinite support whose value is a dyadic rational $p/2^c$ has $\sum_{a \in A} 1/a$ divergent or $\rho(A) > 1$.

The mechanism is a collision: at every positive multiple of L at least $|F|$ support divisors land on the same index, so $f_A \geq |F|$ there, while the exact carry equation $f_A(n) + e_n = k_{n-1} + 2e_{n-1}$ with $k_{n-1} \leq 1$ lets one step absorb at most one unit — so the preceding excess must spike to at least $\lceil (|F|-1)/2 \rceil$, and the spikes recur with density $1/L$. The three steps are formalised by the shifted common-multiple bound, shifted wrap–excess recurrence, and dyadic reciprocal-mass alternative.

Theorem B.11 (carry states are unbounded over infinite supports)**.** *Let A be infinite with value $p/(2^c v)$, $v \geq 1$. Then the positive integer carry state $u(N) = v\, T_{f_A}(c+N)$ is unbounded: for every B there is an N with $u(N) > B$.*

The proof picks $2B+1$ positive support elements; at a common multiple L of all of them, the local inequality $1 + v\,|F| \;\leq\; 2\,u(L-c-1)$ pushes the state past B. The conclusion is the fraction-facing unbounded-state theorem; its local engine is the support-cardinality state bound. So a rational value over an infinite support cannot run on a bounded carry alphabet. That closes finite-state readings of the carry system; it does not close the problem. Nothing in the constraints above prevents an infinite support from satisfying every one of these constraints. The repetend identities and divisor averages used here are classical. We make no claim of mathematical novelty for the coupled reciprocal-mass bounds, collision strengthening, or global unboundedness statement.

B.5 Greedy geometry of the #257 value set

Finally, consider the set of all values a base-2 support series can take. For $n \geq 1$ put $w_n = 1/(2^n - 1)$, let $T(n)$ be the mass strictly after exponent n, and let

$$\mathcal{A} \;=\; \Big\{ \sum_{n \in A} w_n \;:\; A \subseteq \mathbb{N},\; 0 \notin A \Big\},$$

the *Mersenne achievement set*, the exponent 0 being normalised away as analytically invisible. Coding by supports of positive exponents is literally the formal support series (support-value identity). In this language the base-2 #257 statement reads: the rational members of $\mathcal{A}$ are exactly the finite subset sums. Nothing below decides that. Kovač and Tao verify, for every fixed integer $t \geq 2$, that

$$\sum_{\ell > n} \frac{1}{t^\ell - 1} < \frac{1}{t^n - 1},$$

and deduce distinct Lambert subsums and a Cantor set [14]. At $t = 2$ this is the relevant Lambert-series instance of Kakeya's classical strict-tail criterion for subsum sets [9]. The local contribution is the checked greedy criterion, quantitative enclosure, and finite non-membership certificates over that known geometry; it does not attribute those refinements to the cited sources or settle #257.

Theorem B.12 (superincreasing geometry and greedy membership)**.** *For every $n \geq 1$ the weight dominates the whole tail after it, $T(n) < w_n$, with the two-scale gap $w_n - T(n) = \frac{2}{3}4^{-n} + O(8^{-n})$ and explicit valid O-constant 3. Consequently a real x belongs to $\mathcal{A}$ exactly when $x \geq 0$ and every greedy residual of x is at most the remaining tail; and normalised support coding is injective, so each member of $\mathcal{A}$ has a unique support, which the greedy recursion recovers bit by bit.*

Formalised: strict-tail inequality, gap big-O estimate (explicit bound explicit gap bound), greedy survival criterion, and normalised support-value injectivity, transferred to the kernel series at normalised support-series injectivity. The nonnegativity guard in the membership criterion is necessary: without it a negative target would survive every level while lying outside $\mathcal{A}$.

The binary coding also permits the global geometry to be checked directly.

Proposition B.13 (fat-Cantor geometry)**.** *The achievement set $\mathcal{A}$ is compact, perfect, totally disconnected, and nowhere dense. Its Lebesgue measure is exactly one.*

Formalised: measure-one theorem, perfectness, total disconnectedness, and nowhere density. Kovač–Tao supply the strict-tail Cantor-set context cited above. The formalisation records these exact properties without making a novelty or priority claim for them.

Proposition B.14 (exact rational runs and finite non-membership certificates)**.** *The greedy recursion executed in exact rational arithmetic agrees step by step with the real recursion under casting. For a rational target x, write $r_n(x)$ for its exact rational residual after level n and, for a lookahead $\ell \geq 0$, put*

$$\widehat{T}_{n,\ell} = \sum_{k=0}^{\ell-1} w_{n+k+1} + 2w_{n+\ell+1} \in \mathbb{Q}.$$

The sum is empty when $\ell = 0$, and always $T(n) < \widehat{T}_{n,\ell}$. Hence the finite inequality $\widehat{T}_{n,\ell} \leq r_n(x)$ soundly proves $x \notin \mathcal{A}$; its level-one instance shows $3/4 \notin \mathcal{A}$. A rational number already known to lie in $\mathcal{A}$ has computable support bits.

Formalised: rational–real greedy agreement, rational tail-enclosure bound, finite death-certificate soundness, three-quarters nonmembership, and rational-member support-bit equivalence. The certificates are one-sided: survival through any finite depth proves nothing about membership, and no decidability of membership in $\mathcal{A}$ is asserted. Neither the global geometry nor the finite death certificates exclude rational values for either open series.

Appendix consequence. Appendices B.1–B.2 give exact normal forms for rationality; Appendices B.3–B.4 impose necessary asymptotic constraints on any infinite rational support; and Appendix B.5 gives the canonical achievement-set representation and finite nonmembership certificates. None of these necessary conditions is contradictory. Their role is to make the rational side exact and auditable, not to close either open problem.

C Technical reductions for the totient-certificate branch

This appendix expands the supporting mathematics behind Theorem 5.4, Proposition 5.6, and Corollary 5.5. It contains exact cone-flatness and target-interval reformulations, finite residue projections, denominator and bit-lifting formulae, bounded certificate families, and rigorous obstructions to several candidate strategies. Each statement retains its own hypotheses and status. None proves the unbounded certificate supply (5.1).

The dependency order is: exact normal forms in Appendix C.1; finite projections and verified instances in Appendix C.2; fresh-prime and bit-level requirements, including scoped no-go results, in Appendix C.3; and finite determinant and rank routes in Appendix C.4. The first two say exactly what can be checked at one scale; the latter two delimit what would be needed to make those checks unbounded.

C.1 Exact normal forms for the open supply

Cone flatness and completeness. On the cone $\{kM_t : k \geq 1\}$, rationality forces a single fractional constant across the whole cone.

Proposition C.1 (cone flatness)**.** *If S is rational, then for every sufficiently large t there is $\theta_t \in [0, 1)$ such that the fractional part of the tail R_{kM_t} equals θ_t for every $k \geq 1$.*

Formalised: cone flatness under rationality.

The certificates are exact: a certificate exists exactly when the object it certifies is non-integral. Thus the finite computations are machine-checked witnesses rather than numerical experiments.

Proposition C.2 (certificate completeness, expanded form)**.** *For all h, N,*

$$\big(\exists L,\ \mathit{certifiedKill}(h, N, L)\big) \iff R_{N+h} - R_N \notin \mathbb{Z}.$$

Formalised: certificate-completeness equivalence. Conversely, integrality of one positive-shift tail difference forces S to be rational. Thus S is irrational if and only if every positive-shift tail difference is non-integral, equivalently if every such pair admits a finite certificate. These normal forms are formalised at integral-difference rationality criterion, all-differences criterion, and pointwise certificate criterion. Finite certificates witness non-integrality. A joint refuter can interrogate a menu of cone vertices at once and is sharper than any pairwise test: it refutes the single-fractional-part model that rationality predicts on a finite sample of cone vertices (finite cone-nonflatness refuter).

A reduced-denominator finite predicate. Separately, there is a stricter finite condition for an odd reduced denominator u. Its predicate $\mathsf{ReducedDenominatorUnitGapCert}(u, N, K)$ asks that every candidate integer in the associated dyadic tail interval be non-coprime to u. At a prime power $u = p^e$, the condition is equivalent to the exact alternative that the interval has no candidate, or that it has one candidate and p divides that candidate. Thus such an interval has at most one candidate.

Proposition C.3 (prime-power unit-gap ceiling)**.** *For every prime power p^e with $e > 0$, a $\mathit{ReducedDenominatorUnitGapCert}(p^e, N, K)$ has candidate count at most one. The row $(u, N, K) = (3, 3, 5)$ satisfies this predicate, although its corresponding ordinary empty-gap inequality fails.*

Formalised: prime-power unit-gap characterisation, prime-power one-candidate bound, three-window unit-gap certificate, and three-window ordinary-gap failure. This is a finite certificate-level strengthening. We do not establish that rationality of S supplies these predicates at unbounded parameters.

The certificates are non-empty. The initial segment of the required supply is machine-checked, so the reduction is not vacuous. By completeness, each of these is a kernel-checked proof of a specific non-integral tail difference, not a numerical experiment.

Example C.4 (verified finite instances, expanded record)**.** The following are checked by the Lean kernel:

- $\mathsf{certifiedKill}(h, 12, 16)$ for every period $h \in [1, 8]$;

- certified kills for every period $h \in [1, 16]$ at a fixed window;
- diagonal certificates $\mathsf{certifiedKill}(M_t, M_t, L)$ at the 28 scales $t \in \{1, 2, 3, 4, 5, 7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32, 37, 41, 43, 47, 49, 53, 59, 61, 64\}$;
- a tabulated joint cone-vertex certificate.

The four validation classes are formalised, in order, by the small-window certificate family, periods through sixteen, 28 imported diagonal certificates through scale 64, and cone-cell certificate table. These instances show that each layer of the reduction is non-vacuous and, by completeness, exact. They are not evidence that the unbounded supply exists.

The first row of this finite table can be read without any search procedure. Take $(h, N, L) = (1, 12, 16)$. The integer in the definition is

$$\Delta_{1,12,16} = \sum_{j=0}^{15} \big(\varphi(14 + j) - \varphi(13 + j)\big) 2^{15-j} = -143140.$$

Its least nonnegative residue modulo 2^{16} is 53468. The tail allowance is $N + h + L + 2 = 31$, and hence

$$31 < 53468 < 65536 - 31 = 65505.$$

This is precisely $\mathsf{certifiedKill}(1, 12, 16)$. The soundness theorem (certificate soundness) therefore gives $R_{13} - R_{12} \notin \mathbb{Z}$. It is one exact non-integral tail difference, not an assertion that such differences occur at unbounded scales.

An exact target interval. The diagonal calculation also admits a more local normal form. Put $H_t = M_t$ and

$$D_t = R_{2H_t} - R_{H_t}.$$

Separate an exact rational Möbius shadow $Q_t = a_t/d_t$ in lowest terms and define the remaining *foreign defect* by $F_t = D_t - Q_t$. The full target is the residue class of $-a_t$ modulo d_t:

$$\mathsf{FullTargetHit}(t) \quad\Longleftrightarrow\quad \exists k \in \mathbb{Z}, \qquad d_t F_t = -a_t + d_t k.$$

Since $D_t = a_t/d_t + F_t$, this is exactly the condition $D_t \in \mathbb{Z}$; no complementary term is discarded or assigned a conjectural sign.

Proposition C.5 (diagonal pincer decomposition)**.** *For every t, diagonal integrality is equivalent to the foreign diagonal defect meeting the full target. Consequently, full-target avoidance at unbounded scales implies irrationality of S.*

Formalised: full-target equivalence and unbounded target-avoidance criterion. The proposition is an exact reduction; the unbounded avoidance hypothesis is still open.

Two algebraic inputs make this interval test explicit. First, truncate the Möbius-square identity of Proposition D.6 after D terms. The resulting rational quantity gives a Lambert-projected centre $C_{H,D}$ for the scaled diagonal difference. The error is exactly a signed squared-Mersenne tail:

$$\mathsf{scaledDiagonal}(H) - C_{H,D} = c_H \sum_{d>D} \frac{\mu(d)}{(2^d - 1)^2},$$

where c_H is the explicit diagonal coefficient in the formal source. The absolute value of the sum is bounded geometrically by $4/[3(2^{D+1} - 1)^2]$. Thus a rational separation of the centre from every integer by more than the scaled bound proves target avoidance; the argument does not estimate the sign of the tail.

Proposition C.6 (squared-Mersenne enclosure)**.** *The scaled diagonal tail difference differs from its Lambert-projected centre by the squared-Mersenne tail. Separation of the centre from the full target by more than the proved tail bound forces target avoidance.*

Formalised: squared-Mersenne tail identity and Lambert-projected separation criterion.

C.2 Finite projections and verified instances

Finite foreign-residue projection. There is a second finite presentation of the same obstruction. For a cutoff D, split the finite Möbius residue diagonal into the channels $d \le D$ which divide H and the remaining channels. The divisor part is exactly the truncated Möbius shadow, while every retained nondivisor channel is explicit. Above the stable cutoff $2H$, a finite window of omitted channels is bounded by

$$c_H \left(\frac{2}{2^D} + \frac{4}{3\,4^D} \right),$$

where c_H is the diagonal coefficient. Thus, if the actual foreign defect agrees with this projection within the displayed bound, and the projected quantity is farther than that bound from every integer, then the full target is missed.

Proposition C.7 (finite foreign-residue projection)**.** *The finite residue diagonal splits exactly into projected foreign and divisor channels. A controlled projected foreign defect with the stated integer separation cannot meet the full target.*

Formalised: foreign–divisor channel split, omitted-channel bound, and projected-separation criterion. The passage from finite windows to the analytic foreign complement, including the corresponding kernel identity, is not supplied here. Nor is an unbounded family of separated projections; these are parts of the open certificate-supply problem.

Second, the rational Möbius shadow has an exact reduced denominator. If r_t is the squarefree kernel of H_t, $s_t = H_t/r_t$, and $J(r_t)$ is the odd Jordan scalar defined in the algebraic shadow chain, then

$$d_t = \frac{2^{r_t} - 1}{\gcd\big(2^{r_t} - 1,\, s_t |J(r_t)|\big)}.$$

For $t \ge 5$, the product of the Mersenne factors $2^p - 1$ over primes $t/2 < p \le t$ divides d_t; in particular $d_t \ge 2^{t/2}$. This is a denominator theorem for Q_t only. Cancellation by F_t is exactly what the full-target condition retains.

Proposition C.8 (Mersenne-shadow denominator)**.** *At lcm height t, the reduced denominator of the scaled Möbius shadow is given exactly by the formal denominator formula. The upper-half Mersenne product divides it and yields the stated lower bounds.*

Formalised: upper-half Mersenne divisor bound and exact shadow denominator.

Finally, the remaining diagonal error can be represented by a finite integer projection and then by binary suffix data. At depth L, define

$$W_{t,L} = P_L(2H_t) - P_L(H_t), \qquad \rho_{t,L} = W_{t,L} \bmod 2^L,$$

where $P_L(M)$ is the length-L window numerator underlying Definition 5.3. The finite residue certificate is the asymmetric safe interval

$$H_t + L + 2 < \rho_{t,L} < 2^L - (2H_t + L + 2).$$

The two unequal margins are the tail bounds at the two cone vertices H_t and $2H_t$; replacing them by a symmetric slogan would lose part of the checked statement.

Corollary C.9 (finite residue projection)**.** *The diagonal residue certificate is equivalent to its projection inequality, and the projection forces full-target avoidance. Hence an unbounded projection supply, or either stronger adjacent-suffix supply stated in the corresponding theorem, implies irrationality of S.*

Formalised: residue–projection equivalence and unbounded projection criterion. Finite instances of this implication are proved, but no unbounded projection or suffix supply is established.

Primality certificates through $t = 64$. A finite family of Lucas certificates proves the primality facts required by the diagonal-pincer certificates through $t = 64$; for example, representative Lucas primality certificate. This is a finite theorem family, not an unbounded certificate supply or an irrationality proof.

The diagonal certificate at $t = 64$. Using those primality facts, Lean checks the individual diagonal-pincer certificate at $t = 64$, scale-64 diagonal certificate. This is one finite endpoint certificate, not an unbounded supply.

A three-rank odd window. At an analytically admissible odd power-of-two rank, centrality at any one of the canonical odd rank or its next two base-four successors yields the required adjacent-suffix gap condition, three-rank suffix-gap implication. Thus a cofinal three-rank band supply implies irrationality, cofinal three-rank criterion; that supply is not proved.

Affine form of the three-rank condition. With the stated short-correction bounds, the three central-band predicates are equivalent to one three-scale affine escape condition, three-scale affine equivalence. A zero-start affine orbit shows that a uniform correction envelope alone does not force escape at any finite number of ranks, zero-orbit obstruction.

C.3 Fresh-prime and bit-level boundaries

Fresh-prime deficit decomposition. The literal foreign channel also has an endpoint form suited to curvature tests. Retain only the prime support already present at H in the two endpoints $H + s$ and $2H + s$, and write the resulting discrepancy from the actual totient as an endpoint deficit. Each such deficit is nonnegative. The actual diagonal increment then equals the old-prime increment plus the lower deficit minus the upper deficit. The second and five-point curvature operators preserve this exact split; the latter has coefficient row $[-8, 4, 2, 1, 1]$. Consequently, the old curvature margin loses at most the stated adverse weighted deficit.

Proposition C.10 (fresh-prime deficit decomposition)**.** *The endpoint fresh deficits are nonnegative, give an exact decomposition of the foreign channel, and bound the loss from the old-prime curvature margins.*

Formalised: endpoint-deficit nonnegativity, foreign-increment decomposition, and adverse-deficit curvature bound. This supplies the exact correction term for the finite evaluations. It supplies neither a cofinal deficit bound nor an irrationality criterion.

One-bit lifting. The power-of-two signed-margin condition admits a separate exact arithmetic reduction. If a signed difference is already divisible by 2^k, lifting its congruence to 2^{k+1} is equivalent to evenness of the quotient after division by 2^k. Failure of the lift gives the unique translated residue class. In particular, a displayed difference $P - N = 16c$ agrees modulo 32 exactly when c is even; otherwise the residue is the offset class by 16.

Proposition C.11 (one-bit signed-margin lift)**.** *The fifth-bit question after fourth-bit agreement is exactly one cofactor parity test, with the complementary offset class characterising failure.*

Formalised: one-bit divisibility lift, mod-32 cofactor criterion, and offset-16 complement. The centered-state formulation identifies the signed-margin interval condition with failure of this next lift, subject to its stated correction and margin budget: centred-lift equivalence and centred mod-32 equivalence. This is a finite arithmetic equivalence. It does not supply the required cofactors at unbounded scales and gives no irrationality conclusion.

Parity is not a margin certificate. The preceding lift does not settle the reduced signed-margin interval condition. For every positive even modulus and every radius in the stated nontrivial range, each parity class contains both a residue inside the interval and one outside it. The same statement applies to a positive power-of-two modulus. Thus even a fifth-bit computation, by itself, cannot decide the required margin condition.

Proposition C.12 (parity does not decide the reduced margin condition)**.** *Under the explicit radius inequalities, parity does not decide membership in the reduced margin interval, including at a positive power-of-two modulus.*

Formalised: parity obstruction and power-of-two parity obstruction. The result is structural and proves no signed-margin certificate at any particular scale; it excludes parity as a sufficient standalone criterion.

Prime-adjunction no-go theorem. One tempting way to seek an unbounded obstruction is to compare the full target at H, pH, qH, and pqH. The following affine transport law rules out this route before any projection is introduced. For every positive k, the actual diagonal satisfies an affine transport law

$$D_{kH} = Q_k(H) D_H + Z_k(H), \qquad Q_k(H) \in \mathbb{N}, \quad Z_k(H) \in \mathbb{Z}.$$

Hence integrality at H propagates to every multiple, and the four target hits at the prime-adjunction diamond are equivalent to the single hit at its root (prime-adjunction diamond collapse). The corresponding transport of the complete foreign defect has zero diamond curvature (foreign-correction flatness). Thus explicit-shadow fibres alone cannot create a holonomy obstruction. A future positive route would need a separately defined projection together with a theorem controlling what that projection discards. This is a proved obstruction to one strategy, not progress on the unbounded-supply proposition.

Proposition C.13 (scalar-localisation obstruction)**.** *Let $x \in \mathbb{Q}$ and $c \in \mathbb{Z}$. If $H \mid \operatorname{den}(x)$ and the reduced denominator of cx divides H, then*

$$\frac{\operatorname{den}(x)}{H} \mid |c|.$$

Formalised: complementary-denominator divisibility and scalar-localisation identity. Thus a scalar localisation can move a complementary denominator factor into its coefficient, but cannot erase it. This rules out a scalar-denominator shortcut of that form. It neither proves full-target avoidance nor supplies any certificates, so it gives no irrationality criterion for S.

Proposition C.14 (square-CRT correction suppression)**.** *Let E be a finite family indexed by distinct primes p_i, with prescribed anchors A_i and residues a_i. There is a bounded common base n such that, for each $i \in E$ and each shift h with $p_i \nmid a_i + h$, one has*

$$n = A_i + p_i(a_i + p_i t) \quad \text{and} \quad \varphi(n + p_i h - A_i) = (p_i - 1)\varphi(a_i + p_i t + h)$$

for a suitable $t \in \mathbb{N}$. In particular, if every p_i exceeds a fixed horizon J, the choice $a_i = 0$ works simultaneously for all $1 \le h \le J$.

Formalised: finite clean totient family and clean horizon family. The finite mechanism does not force a cube coefficient to be nonzero: there is a checked clean two-step cube with both displayed coefficients zero (vanishing clean block) and a separate checked clean cube with second coefficient -4 (nonzero clean block). Thus the result gives neither an unbounded certificate supply nor an irrationality criterion for S.

C.4 Finite determinant and rank routes

Proposition C.15 (signed dyadic-moment algebra)**.** *For matrices $M : \iota \times \kappa \to R$ and $N : \kappa \times \iota \to R$ over a commutative ring, the determinant of MN is the sum over all maps $p : \iota \to \kappa$ of the determinant of the corresponding selected-column matrix times the diagonal product $\prod_i N_{p(i),i}$. Separately, if a selected integer numerator is odd and its dyadic exponent is strictly larger than every other selected exponent, then the common numerator after dyadic denominator clearing is odd and hence nonzero.*

Formalised: rectangular determinant expansion, dominant dyadic parity, and dominant dyadic nonvanishing. These are finite algebraic inputs to a signed-moment route. They do not show that any actual totient Hankel determinant is nonzero, and they give no irrationality criterion for S.

Proposition C.16 (dyadic-kernel rank criterion)**.** *For every fixed level e, a separated nonzero evaluation minor of the canonical dyadic totient-kernel family gives its span dimension $2^e + 1$. Every even residue channel is exactly a scalar multiple of its odd-core channel. Moreover, a compressed-adjoint certificate whose nonzero multiple is strictly smaller in absolute value than its positive modulus is inconsistent.*

Formalised: separated-minor rank criterion, even-residue reduction, and compressed-adjoint contradiction. What is missing is an all-level construction of the separated minors, for example by a CRT–Dirichlet argument. The proposition does not provide those certificates for every e, and it gives no irrationality criterion for S.

Proposition C.17 (residual-gauge obstruction for monomial minors)**.** *Let a monomial matrix have a common nonzero residual weight in each column. The residual weights act by a right diagonal gauge, and therefore preserve determinant nonvanishing. If one exponent is 1, the inverse-phase locked gauge makes that row identically 1 while preserving a nonzero determinant whenever the coefficient-side determinant is nonzero. If the residual is allowed to depend on both row and column, it reconstructs a consecutive-power matrix whose zeroth row is identically 1.*

Formalised: residual-gauge determinant equivalence, locked-reconstruction nonvanishing, and reconstructed zero row. Thus a residual-blind rank, determinant, or conditioning criterion alone cannot exclude the target. The proposition does not exclude a criterion that also uses an arithmetic coupling identity, and it gives no irrationality criterion for S.

Appendix consequence. The first subsection gives exact normal forms for the same unbounded supply as Theorem 5.4; the second turns those forms into finite projections and checked instances. The last two subsections identify additional arithmetic inputs and prove that several weaker parity, prime-adjunction, scalar, and residual-blind rank criteria cannot provide them. No layer changes the quantifier: a positive proof must still produce certificates beyond every threshold.

D Lambert, probability, and finite-algebraic complements

This appendix records results that enlarge the surrounding Lambert-series picture without advancing either unbounded frontier in Section 8. They include periodic-weight theorems, finite denominator obstructions, the Möbius-square identity, a squared Lambert calculus for gcd moments, an exact Stern–Brocot cylinder law, and sharp Fibonacci–continuant stability for alternating runs.

D.1 Additional Lambert, probability, and finite-algebraic structure

Periodic weights. For $L_b(\gamma) = \sum_{a\geq 1} \gamma(a)/(b^a - 1)$, the checked endpoints are:

Theorem D.1 (eventually-periodic nonnegative weights)**.** *An eventually-periodic nonnegative rational weight with a positive periodic value gives an irrational series.*

Formalised: eventually-periodic nonnegative criterion. This is contained in a broader theorem of Luca–Tachiya [13].

Theorem D.2 (periodic signed-weight dichotomy)**.** *For a periodic integer weight, the series is irrational or terminating in base b.*

Formalised: periodic signed-weight dichotomy.

Proposition D.3 (residue-blind obstruction)**.** *For the period-four weight $1, 0, -1, 0$, the divisor coefficient vanishes on $n \equiv 3 \pmod 4$.*

Formalised: period-four coefficient obstruction.

Finite algebraic boundaries. Put $a_n = (\varphi * \mu)(n)/n$. If an integer D clears these coordinates through N, the exact obstruction is:

Proposition D.4 (primitive-coordinate finite-jet obstruction)**.** *D is divisible by an explicit two-tier primorial T_N; hence no fixed D clears every a_n.*

Formalised: two-tier primorial obstruction.

For $d \geq 1$, define the pointed periodic atom

$$\omega_d(N) = \frac{2^{N \bmod d}}{2^d - 1}.$$

It satisfies the exact carry law

$$2\omega_d(N) - \omega_d(N+1) = \mathbf{1}_{d \mid N+1}.$$

Proposition D.5 (finite Mersenne-atom determinant boundary)**.** *Fix $m \geq 1$, conductors $d_j \geq 1$, indices $n_{ij} \geq 0$, and integer column weights w_j, and set*

$$W_{ij} = w_j \omega_{d_j}(n_{ij}), \qquad U_{ij} = 2^{n_{ij} \bmod d_j}.$$

Then

$$\left(\prod_{j=1}^{m} (2^{d_j} - 1) \right) \det W = \left(\prod_{j=1}^{m} w_j \right) \det U \in \mathbb{Z}.$$

If the integer on the right is nonzero, then the absolute value of the cleared determinant on the left is at least 1.

Formalised: Mersenne-atom carry law, determinant-clearing identity, and cleared-determinant lower bound. Thus finite Mersenne-atom determinants recover exactly their explicit column denominators; the clearing creates no additional height surplus. These are scoped finite-denominator obstructions, not irrationality criteria for S.

Proposition D.6 (the Möbius-square lens)**.**

$$S = \tfrac{1}{2} + \sum_{d \geq 1} \frac{\mu(d)}{(2^d - 1)^2}.$$

Formalised: Möbius-square identity. Writing $T = \sum_{d\geq 1} \mu(d)/(2^d - 1)^2$, irrationality of S is equivalent to irrationality of T, since $S = \frac{1}{2} + T$. The factor $1/(2^d - 1)^2$ is the probability that d divides two independent fair-coin waiting times (Section 5.2). The signed constant, the coprime-pair mass, and the tail differences of Section 5.3 are three readings of the same quantity.

Squared Lambert calculus. For $L_2(f)=\sum_{d\ge 1} f(d)/(2^d-1)^2$, the denominator is the probability that d divides two independent geometric waiting times. The formal transfer theorem, squared Lambert transfer, therefore gives a divisor calculus for their gcd.

Proposition D.7 (squared Lambert ladder)**.** *At $r=1/2$, the three rows $f=\mu,\mathbf{1},\varphi$ yield respectively*

$$L_2(\mu)=S-\tfrac{1}{2},\qquad L_2(\mathbf{1})=\sum_{n\ge 1}\frac{\sigma(n)-\tau(n)}{2^n},\qquad L_2(\varphi)=\sum_{n\ge 1}\frac{P(n)-n}{2^n},$$

where P is Pillai's gcd-sum function.

Formalised: totient gcd-moment identity. Postelmans–Van Assche imply irrationality of the middle row [12]; this does not transfer to the Möbius row, which is exactly the open constant $S-\frac{1}{2}$.

Reduced slopes and Stern–Brocot cylinders. The fair-coin pair has a particularly rigid distribution after division by its gcd. For positive coprime a,b, put

$$\nu(a,b)=\frac{1}{2^{a+b}-1},\qquad M(a,b)=\frac{1}{(2^a-1)(2^b-1)}.$$

The first quantity is the probability that the reduced direction is (a,b); the second is the mass of the Stern–Brocot cylinder rooted at (a,b). Let $M_d(a,b)$ be the total stop mass in the first d generations below that root, where a node (u,v) has stop mass $(2^{u+v}-1)^{-1}$ and children $(u+v,v)$, $(u,u+v)$.

Theorem D.8 (reduced-direction and cylinder law)**.** *The reduced directions form a probability distribution,*

$$\sum_{\substack{a,b\ge 1\\(a,b)=1}}\nu(a,b)=1.$$

At every positive node,

$$M(a,b)=\frac{1}{2^{a+b}-1}+M(a+b,b)+M(a,a+b),$$

and, uniformly in a,b and d,

$$0\le M(a,b)-M_d(a,b)\le\left(\frac{2}{3}\right)^d M(a,b).$$

Consequently $M_d(a,b)\to M(a,b)$; at the root $M(1,1)=1$.

Formalised: reduced-direction mass one, cylinder mass recursion, depth-d remainder bound, and cylinder convergence. The recursion is the elementary rational identity obtained after setting $A=2^a$, $B=2^b$; the quantitative point is that the two children retain at most two thirds of their parent's mass.

Alternating-run stability. Write an alternating Stern–Brocot word as r nonempty runs of lengths $n_i=1+e_i$, with $e_i\ge 0$. Starting with $P_\varnothing=(1,1)$, define

$$P_{(n_1,\dots,n_r)}=(n_1A+B,A)\quad\text{when}\quad P_{(n_2,\dots,n_r)}=(A,B),\qquad H(n_1,\dots,n_r)=P_1+P_2.$$

Thus H is the arithmetic height of the primitive pair reached by the word. Let $F_0=0,F_1=1$.

Theorem D.9 (Fibonacci–continuant stability)**.** *Every r-run word satisfies*

$$H(1+e_1,\dots,1+e_r) \geq F_{r+3} + F_{r+1}\sum_{i=1}^{r} e_i,$$

and $H(1,\dots,1) = F_{r+3}$. A defect confined to position $i \in \{0,\dots,r-1\}$ has the exact size

$$H(\underbrace{1,\dots,1}_{i},1+e,\underbrace{1,\dots,1}_{r-i-1}) = F_{r+3} + F_{i+2}F_{r-i+1}e.$$

More generally, the checked recurrence gives the full positive multiaffine continuant defect. The two orientations with one nonempty run have total mass $1/2$, while the natural denominator exponent on the all-unit spine is

$$\sum_{j=0}^{r-1} F_{j+2} = F_{r+3} - 2.$$

Formalised: Fibonacci height lower bound, multiaffine run-height formula, aggregate defect lower bound, one-site defect identity, one-run mass, and natural denominator exponent.

The Lambert and Möbius identities used above are classical [6, 7, 8], and the mediant language goes back to Farey [10]. Postelmans–Van Assche supply the cited irrationality of the constant-weight squared-Lambert row, not these cylinder or run statements. We claim the displayed Lean-checked identities and estimates, not priority for their assembled package. In particular, the exact equality of the Fibonacci height and denominator scales is a boundary, not an irrationality argument: no theorem here proves that a run tail survives denominator clearing.

Appendix consequence. This appendix contains three different statuses. The periodic-weight irrationality theorems are unconditional but lie within the cited classical and modern theory; the Möbius-square, gcd-moment, cylinder, and continuant statements are unconditional structural results; and the finite-algebraic propositions delimit strategies without producing an irrationality criterion. None supplies the unbounded certificate or half-membership input isolated in Section 8.

E Guide to the formal sources

The paper is meant to be read as mathematics. The blue mathematical phrases following formalised statements are optional links to the exact checked theorems in the archived source revision. Exact declaration names remain in the machine-readable audit layer.

For a first reading, the main landmarks are Theorem 4.1 for the full-support irrationality theorem, Theorem 5.1 for the totient denominator exclusion, Theorem 5.4 for the exact tail-certificate reduction, and Proposition B.13 for the geometry of the Mersenne achievement set. For the strongest independent structure beyond those two proof spines, see Theorems D.8 and D.9. The public repository contains a complete searchable cross-reference for readers who need declaration-level detail.

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